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Chapter 1: Orienting Yourself - The Use of Coordinates

This chapter teaches you how to describe the exact position of any point on a flat surface using nothing but two numbers. That sounds small, but it is the foundation of coordinate geometry, which you will use in every higher class. The uploaded PDF covers the 2-D Cartesian system (axes, origin, ordered pairs, quadrants, plotting), distances parallel to the axes, the distance formula built on the Baudhayana-Pythagoras Theorem, reflections in the axes, midpoint and trisection ideas, the collinearity test, and points on a circle centred at the origin. In this practice book we push every one of those ideas one level higher, with harder numbers, missing-value reasoning, and proof-style thinking.

1.1 Complete Concept Cheat Sheet (read this before any question)

1.1.1 The coordinate system and its parts

What is it? A coordinate system is a grid-based framework that locates every point in a plane using an ordered pair of numbers. Two perpendicular number lines are drawn: the horizontal one is the **x-axis** and the vertical one is the **y-axis**. They meet at the **origin O = (0, 0)**. Values to the right of O on the x-axis and above O on the y-axis are positive; values to the left and below are negative.

When do I use it? Whenever a question gives you points, positions on a map or a room layout, distances between locations, or asks you to 'plot', 'locate', 'identify the quadrant', or 'find the distance between points'.

How do I recognise it? Any mention of (x, y) pairs, axes, quadrants, graph paper, or a floor plan with walls along two directions. **What does each part mean?** In the pair (x, y), the first number x is the **perpendicular distance from the y-axis** (measured along the x-axis direction), and the second number y is the **perpendicular distance from the x-axis**. The pair is called an *ordered pair* because the order is fixed: (3, 5) and (5, 3) are different points. Logically, you can understand it as 'walk x units horizontally, then y units vertically from O'.

Tiny example. The point (2, 3) is 2 units right of O and 3 units up. The point (-2, 3) is 2 units LEFT and 3 up. Same distances, different signs, different points entirely.

1.1.2 Points on the axes and the quadrant sign rules

- A point on the x-axis always looks like (x, 0). Its y-coordinate is 0 because it has zero height above the x-axis. Positive x means right of O, negative x means left of O.
- A point on the y-axis always looks like (0, y). Its x-coordinate is 0. Positive y means above O, negative y means below O.
- The origin (0, 0) lies on BOTH axes; it belongs to no quadrant.
- Quadrant I: (+, +). Quadrant II: (-, +). Quadrant III: (-, -). Quadrant IV: (+, -). Memorise the cycle: starting from I and going anticlockwise the signs are ++, -+, --, +-. A quick memory trick: signs of y follow the C-shape of Roman numerals I, II, III, IV written anticlockwise; signs of x are + in I, - in II, - in III, + in IV.
- Any point with x = 0 and y = 0 must be the origin. Any point with both coordinates non-zero cannot lie on either axis.

> **Identify-the-method trick**
> See a question that says 'on the x-axis' or 'on the y-axis'? Instantly write the fixed coordinate: (something, 0) or (0, something). This single habit unlocks a whole family of missing-coordinate questions.

1.1.3 Distances parallel to the axes

What is it? If two points share the same y-coordinate, the segment joining them is horizontal, and the distance is simply the absolute gap between the x-values: distance = $|x_2 - x_1|$. If two points share the same x-coordinate, the segment is vertical and the distance is $|y_2 - y_1|$.

When do I use it? Whenever two given points have one identical coordinate - check this FIRST before reaching for the distance formula. **How do I recognise it?** Same second number means a horizontal move (like walking along a corridor east-west); same first number means a vertical move. **Each term:** the absolute value bars guarantee a positive (non-negative) length; order of subtraction does not matter.

Tiny example. Distance between (2, 5) and (9, 5) is $|9 - 2| = 7$. Distance between (4, -3) and (4, 6) is $|6 - (-3)| = 9$. Note the classic sign trap: $6 - (-3) = 9$, not 3.

1.1.4 The distance formula (the star of the chapter)

What is it? For any two points $A(x_1, y_1)$ and $D(x_2, y_2)$, the distance is $AD = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$. **Where does it come from?** Drop a right-angled helper triangle: the horizontal leg is $|x_2 - x_1|$ and the vertical leg is $|y_2 - y_1|$. The segment AD is the hypotenuse, so the Baudhayana-Pythagoras Theorem ($leg^2 + leg^2 = hypotenuse^2$) gives the formula. It is not a magic rule - it is Pythagoras wearing coordinates.

When do I use it? When the two points differ in BOTH coordinates. **How do I recognise it?** Words like 'find the length of AB', 'how far apart', 'show that the triangle is isosceles', 'find the perimeter' with slanted sides. **Each term:** subtract corresponding coordinates, square each difference (squares kill any minus sign, so the order of the points does not matter), add, then take the positive square root.

Can it be derived logically? Yes - see the helper-triangle picture in Figure 2. **Tiny example.** From (1, 2) to (4, 6): differences are 3 and 4, so distance = $\sqrt{(9 + 16)} = \sqrt{25} = 5$. Whenever the squared sum is a perfect square you get a clean integer - examiners love the triples 3-4-5, 5-12-13, 6-8-10, 8-15-17, 9-12-15, 7-24-25.

- **Method 1 (helper triangle):** sketch, mark legs |dx| and |dy|, apply Pythagoras. Best for learning and for verifying.
- **Method 2 (formula directly):** substitute into $\sqrt{(dx^2 + dy^2)}$. Best for speed once confident.
- **Method 3 (shift to origin):** subtract the first point from both - distance is unchanged - then compute from (0, 0). Great mental shortcut: distance of (x, y) from origin is $\sqrt{(x^2 + y^2)}$.

> **Common confusion**
> Students often compute $\sqrt{(dx)} + \sqrt{(dy)}$ or $dx + dy$ instead of $\sqrt{(dx^2 + dy^2)}$. Remember: square EACH difference first, THEN add, THEN take one single square root at the very end. Never take a root midway.

1.1.5 Reflections in the axes

What is it? Reflecting a point in the x-axis flips its height: (x, y) becomes (x, -y). Reflecting in the y-axis flips its horizontal position: (x, y) becomes (-x, y). Reflecting in BOTH axes (equivalent to a half-turn about O)

gives (-x, -y). **Why only one sign changes:** a mirror in the x-axis changes only how far up or down you are; your sideways position stays the same.

Key property: reflection preserves distances. If A reflects to A' and B reflects to B', then AB = A'B'. The PDF proves this by computing both distances - the squares of the coordinate differences come out identical. Use this to check your reflection answers. **Tiny example.** A(2, 3) reflects in the x-axis to A'(2, -3); distance from A to any point is unchanged, e.g. if B(6, 3) then AB = 4 and A'B = $\sqrt{(16 + 36)} = \sqrt{52}$? No wait - check: B(6,3) reflects to B'(6,-3), and A'B' = |6-2| = 4 = AB. Reflection maps the whole segment, not just one endpoint.

1.1.6 Midpoint and equal-distance ideas

What is it? The midpoint of a segment splits it into two equal halves. The PDF discovers the connection experimentally: if M(m_1, m_2) is the midpoint of S and T, then each coordinate of M is the average of the corresponding coordinates: $m_1 = (x_S + x_T)/2$ and $m_2 = (y_S + y_T)/2$. You can verify with the distance formula: distances from M to S and from M to T come out equal. **Reverse use:** if you know one endpoint A and the midpoint M, the other endpoint is B = ($2m_1 - x_A$, $2m_2 - y_A$) - just undo the averaging. **Trisection:** points P and Q that cut AB into three equal parts satisfy P = A + (B - A)/3 and Q = A + 2(B - A)/3; the midpoint of P and Q equals the midpoint of AB, which is a lovely self-check.

When do I use the average idea? Keywords: 'midpoint', 'bisects', 'equally spaced', 'middle point of the segment'. **Tiny example.** Midpoint of (-4, 1) and (6, 5) is ((-4 + 6)/2, (1 + 5)/2) = (1, 3).

1.1.7 Collinearity test without plotting

What is it? Three points are collinear (lie on one straight line) if one of the three distances equals the sum of the other two. Concretely, compute AB, BC, AC and check whether the largest equals the sum of the other two. **Why it works:** if B truly sits on the straight path from A to C, then walking A to B to C covers exactly the straight distance AC - no detour is possible on a straight line (triangle inequality with equality).

When do I use it? Questions like 'are these points on the same straight line?' or 'check without plotting'. **Trap:** the check must be done with EXACT values. If the distances involve roots, compare exact squared forms or keep radicals; do not round early. **Tiny example.** (0,0), (3,4), (6,8): AB = 5, BC = 5, AC = 10, and 5 + 5 = 10, so collinear.

1.1.8 Points on a circle centred at the origin

What is it? A circle with centre O(0, 0) and radius r is the set of points whose distance from O equals exactly r. So a point (x, y) lies ON the circle if $x^2 + y^2 = r^2$, INSIDE if $x^2 + y^2$ is less than r^2, and OUTSIDE if it is greater. This is just the distance-from-origin shortcut applied with an equality/inequality. **How do I recognise it?** 'Show that these points lie on a circle with centre the origin - find the radius' or 'is the point inside, on, or outside the circle?'. **Tiny example.** (1, -8), (-4, 7), (-7, -4): squares give 65, 65, 65 - all equal, so all three lie on the circle $x^2 + y^2 = 65$, radius $\sqrt{65}$.

1.1.9 Exam tips for this chapter

1. Before any distance computation, CHECK whether a coordinate is shared - the parallel-to-axes shortcut saves time and avoids errors.
2. Keep the ordered-pair order sacred. (x, y) and (y, x) coincide only when x = y.

3. Squaring removes minus signs, so $(x_2 - x_1)^2 = (x_1 - x_2)^2$. Do not waste time worrying about subtraction order.
4. Memorise Pythagorean triples; they turn ugly roots into integers.
5. For 'inside/on/outside circle' questions, compare SQUARES of distances - never square roots.
6. After finding a reflection, quickly verify one preserved distance.
7. In multi-part geometry questions, draw a rough sketch even if the question says 'without plotting' - your sketch guides the algebra; the algebra is what you present.

1.2 Question Types / Concept Map for Chapter 1

Almost every coordinate question in exams falls into one of the following types. Use this map to label what you are looking at before solving - half the battle is knowing WHICH type of fight it is.

Type	Name	How to recognise it in the question	Core method
T1	Direct identification	Given coordinates, find quadrant / axis / sign pattern	Sign rules of the four quadrants
T2	Plotting and reading	Mark points, read positions from a room/map figure	Ordered pair = walk x then y
T3	Axis-parallel distance	Two points share one coordinate	|dx| or |dy| shortcut
T4	Distance formula	Two points, both coordinates differ	$\sqrt{(dx^2 + dy^2)}$
T5	Reverse / missing value	Distance given, one coordinate unknown (find k)	Set up equation, solve, check BOTH roots
T6	Triangle / quadrilateral classification	Show isosceles / right / square / rhombus / parallelogram	Compute all sides (and diagonals), compare
T7	Reflection reasoning	Image of a point in x-axis / y-axis / both	Flip the relevant sign; verify preserved distance
T8	Midpoint / trisection	'M is the midpoint', 'points of trisection', 'bisects'	Average coordinates; reverse-average for missing endpoint
T9	Collinearity test	'On the same line?', 'without plotting'	Largest distance = sum of other two (exact values)
T10	Circle membership	'lie on circle with centre origin', inside/outside check	Compare $x^2 + y^2$ with r^2
T11	Applied grid problems	City streets, screen pixels, room maps	Model positions as coordinates, then use T3/T4
T12	Perimeter / area from coordinates	Vertices given, find perimeter or area	Distance formula on each side; 1/2 base x height for right triangles
T13	Conceptual trap questions	Order of pair, meaning of coordinates, statements to justify	Reason from definitions, not memory

Type	Name	How to recognise it in the question	Core method
T14	Multi-concept HOTS	Combines distance + midpoint + circle or reflection chains	Break into labelled mini-steps, verify at each junction

Distribution of the 50 practice questions below: T1-T3 feed Q1-Q10, T4-T5 feed Q11-Q20, T6-T9 feed Q21-Q35, T10-T13 feed Q31-Q40, and T14 dominates Q41-Q50.

1.3 Practice Questions (Q1 - Q50) with Full Step-by-Step Solutions

Difficulty climbs in five bands: Q1-Q10 strong basics, Q11-Q20 moderate, Q21-Q30 difficult, Q31-Q40 high-level, Q41-Q50 very challenging (Olympiad-lite). Every question carries the concept tag so you can trace it back to the cheat sheet.

Q1 · Basic · Concept: Quadrant sign rules (T1)

Q1. Name the quadrant (or axis) in which each point lies, and state the sign pattern: A(7, -9), B(-6, 2), C(-8, -11), D(9, 4), E(0, -3), F(-12, 0).

SOLUTION

Step 1: Given: six points with their coordinates. Needed: location of each point.

Step 2: Recall the sign pattern: Quadrant I (+, +), Quadrant II (-, +), Quadrant III (-, -), Quadrant IV (+, -). Points with x = 0 lie on the y-axis; points with y = 0 lie on the x-axis.

Step 3: A(7, -9): x is positive, y is negative, so A is in Quadrant IV with signs (+, -).

Step 4: B(-6, 2): x negative, y positive, so Quadrant II with signs (-, +).

Step 5: C(-8, -11): both negative, so Quadrant III with signs (-, -).

Step 6: D(9, 4): both positive, so Quadrant I with signs (+, +).

Step 7: E(0, -3): x-coordinate is 0, so E lies on the y-axis, 3 units below the origin, in no quadrant.

Step 8: F(-12, 0): y-coordinate is 0, so F lies on the x-axis, 12 units left of the origin.

Hinglish: *Beta, yahan sirf signs dekhne hain. Pehle number jo hai wo x ki duniya hai - positive matlab right side, negative matlab left. Doosra number y hai - positive matlab upar, negative matlab neeche. A mein x positive, y negative, matlab right aur neeche = Quadrant IV. Bas ye pattern yaad rakho: I mein dono plus, II mein x minus y plus, III mein dono minus, IV mein x plus y minus. Aur agar koi coordinate 0 hai toh point axis par baitha hai, quadrant mein nahi.*

Answer: A in Q IV (+, -); B in Q II (-, +); C in Q III (-, -); D in Q I (+, +); E on the y-axis below O; F on the x-axis left of O.

Q2 · Basic · Concept: Points on the axes (T1/T2)

Q2. Write the coordinates of (i) the point on the x-axis that is 6 units to the left of the origin, (ii) the point on the y-axis that is 4.5 units below the origin, (iii) the point where the two axes meet.

SOLUTION

Step 1: Given: descriptions along the axes. Needed: coordinates.

Step 2: A point on the x-axis has the form (x, 0); 'left of the origin' means the x-value is negative.

Step 3: So 6 units left of O gives (-6, 0).

Step 4: A point on the y-axis has the form (0, y); 'below the origin' means the y-value is negative.

Step 5: So 4.5 units below O gives (0, -4.5).

Step 6: The axes meet at the origin, whose coordinates are (0, 0).

» **Why this step?** We fix the zero coordinate FIRST because 'on the axis' forces it; then only the sign and the size of the other coordinate remain.

Hinglish: *X-axis par rehne wale har point ka second number hamesha 0 hota hai - ye pakka rule hai. Left jaana hai matlab negative. Isliye (-6, 0). Y-axis par rehne wale ka pehla number 0 hota hai, aur neeche matlab negative y, isliye (0, -4.5). Origin dono axis ka intersection hai, wahan dono zero - (0, 0). Sign ka khayal rakho, bas yahi galti hoti hai.*

Answer: (i) (-6, 0) (ii) (0, -4.5) (iii) O(0, 0)

Q3 · Basic · Concept: Meaning of coordinates (T13)

Q3. For the point P(5, -7), state (i) its perpendicular distance from the y-axis, (ii) its perpendicular distance from the x-axis, (iii) which side of each axis it lies on.

SOLUTION

Step 1: Given: P(5, -7). Needed: distances from the two axes and directions.

Step 2: By definition, the x-coordinate is the perpendicular distance from the y-axis, measured along the x-direction.

Step 3: So the distance from the y-axis is 5 units, and since 5 is positive, P lies to the right of the y-axis.

Step 4: The y-coordinate is the perpendicular distance from the x-axis.

Step 5: So the distance from the x-axis is 7 units, and since -7 is negative, P lies below the x-axis.

Step 6: Hence P sits in Quadrant IV, 5 units right and 7 units down from the origin.

» **Why this step?** Distances are always written positive; the SIGN of the coordinate only tells you the side. Never say the distance is -7.

Hinglish: *Bahut important concept: x-coordinate batata hai y-axis se doori, aur y-coordinate batata hai x-axis se doori. Students yahan ulta kar dete hain! P(5, -7) ka matlab: y-axis se 5 unit door (right side), x-axis se 7 unit door (neeche, kyunki minus hai). Doori kabhi negative nahi hoti - minus sirf direction batata hai.*

Answer: (i) 5 units, right of the y-axis (ii) 7 units, below the x-axis (iii) Quadrant IV

Q4 · Basic · Concept: Ordered pair order matters (T13)

Q4. Are the points (3, 5) and (5, 3) the same point? Justify. For what values of a and b does the point (a, b) coincide with the point (b, a)?

SOLUTION

Step 1: Given: two ordered pairs. Needed: whether they name the same location.

Step 2: The point (3, 5) means 3 units along the x-direction and 5 units along the y-direction; (5, 3) means 5 along x and 3 along y.

Step 3: These are different walks from the origin, so they land at different places: (3, 5) is above-left of (5, 3). They coincide only if the two coordinates are equal.

Step 4: Setting (a, b) = (b, a) gives a = b and b = a - the same single condition a = b.

Step 5: So for any value with a = b (for example (4, 4)), the two pairs name one and the same point on the line y = x.

» **Why this step?** The justification is definitional, not visual: ordered pairs are equal exactly when they agree coordinate by coordinate.

Hinglish: *(3, 5) aur (5, 3) bilkul alag points hain kyunki ordered pair mein ORDER matter karta hai - pehle x ki baat, phir y ki. Dono tabhi ek hi point banenge jab x aur y barabar hon, yaani a = b. Tab dono us line par baithte hain jahan har point ka x aur y same hota hai (line y = x). Exam mein ye 'true/false with reason' type bahut aata hai.*

Answer: No, (3, 5) and (5, 3) are different points; (a, b) = (b, a) exactly when a = b.

Q5 · Basic · Concept: Plotting and locating (T2)

Q5. Without actually drawing, describe the position of P(-3.5, 2.5) and Q(2.5, -3.5). Are P and Q reflections of each other in either axis? Give a reason.

SOLUTION

Step 1: Given: two points with fractional coordinates. Needed: rough location and reflection relation.

Step 2: P has x = -3.5 (left of the y-axis) and y = +2.5 (above the x-axis), so P sits in Quadrant II.

Step 3: Q has x = +2.5 (right) and y = -3.5 (below), so Q sits in Quadrant IV.

Step 4: Reflecting P in the x-axis would give (-3.5, -2.5); reflecting P in the y-axis would give (3.5, 2.5). Neither equals Q(2.5, -3.5).

Step 5: Also notice P and Q have swapped magnitudes: 3.5 with 2.5 exchanged places. A reflection in an axis flips ONE sign and keeps both magnitudes in place, so P and Q are not axis reflections of each other.

Hinglish: *P left-upar wale hisse (Quadrant II) mein hai, Q right-neeche (Quadrant IV) mein. Reflection ka rule yaad karo: x-axis mein reflect karne par sirf y ka sign ulta hota hai, y-axis mein sirf x ka. Yahan signs ka pattern milta nahi (P ko flip karo toh (−3.5, −2.5) ya (3.5, 2.5) banta hai, Q toh (2.5, −3.5) hai). Aur dhyan do - P aur Q mein numbers ne apas mein jagah bhi badal li hai (3.5 aur 2.5 swap). Reflection mein swapping kabhi nahi hoti. Isliye ye reflections nahi hain.*

Answer: P is in Quadrant II, Q is in Quadrant IV; they are NOT reflections of each other in either axis (a reflection flips exactly one sign and swaps no coordinates).

Q6 · Basic · Concept: Axis-parallel distances (T3)

Q6. Find the distance between (i) A(2, 0) and B(9, 0), (ii) C(0, -4) and D(0, 6), (iii) E(-5, 3) and F(4, 3).

SOLUTION

Step 1: Given: three pairs of points. Needed: distances.

Step 2: Pair (i): both points lie on the x-axis (y = 0 for both), so use the horizontal shortcut |x2 - x1| = |9 - 2| = 7.

Step 3: Pair (ii): both on the y-axis, so distance = |y2 - y1| = |6 - (-4)| = |6 + 4| = 10.

Step 4: Pair (iii): both have y = 3, a horizontal segment: distance = |4 - (-5)| = |4 + 5| = 9.

Step 5: In every case the shared coordinate confirms the segment is parallel to one axis, so the single-difference shortcut is valid - no square roots needed.

» Why this step? Checking the shared coordinate FIRST is the habit that separates fast, accurate students from slow ones.

Hinglish: *Jab do points mein se koi ek coordinate same ho, toh seedha difference lo - formula ki zaroorat hi nahi. (ii) mein dhyan do: 6 - (-4) = 6 + 4 = 10. Minus ke saath minus mila toh plus ban gaya. Ye sign wala jhatka sabse common galti hai, isliye maine absolute value ke andar rakha hai. (iii) bhi same khel hai: 4 - (-5) = 9.*

Answer: (i) 7 units (ii) 10 units (iii) 9 units

Q7 · Basic · Concept: Perimeter from coordinates (T12/T3)

Q7. The corners of a rectangular room are O(0, 0), A(8, 0), B(8, 5) and C(0, 5). Find the perimeter and state which walls are parallel to which axes.

SOLUTION

Step 1: Given: four corners in order. Needed: perimeter and axis directions of walls.

Step 2: OA joins (0, 0) to (8, 0): both y-values are 0, so OA is horizontal with length |8 - 0| = 8.

Step 3: AB joins (8, 0) to (8, 5): both x-values are 8, so AB is vertical with length |5 - 0| = 5.

Step 4: BC joins (8, 5) to (0, 5): horizontal, length |8 - 0| = 8. CO joins (0, 5) to (0, 0): vertical, length 5.

Step 5: Perimeter = 8 + 5 + 8 + 5 = 26 units.

Step 6: OA and BC run along the x-direction (parallel to the x-axis); AB and CO run along the y-direction (parallel to the y-axis).

Hinglish: *Rectangle ke opposite sides barabar hoti hain, isliye do sides hi kaafi hain: ek horizontal 8, ek vertical 5. Perimeter = 2(8 + 5) = 26. Dekho, coordinates same hone se hi pata chal gaya ki kaun si deewar kis axis ke parallel hai - y same hai toh horizontal wall, x same hai toh vertical wall. Ye floor-plan wale questions mein bahut kaam aata hai.*

Answer: Perimeter = 26 units; OA and BC are parallel to the x-axis, AB and CO are parallel to the y-axis.

Q8 · Basic · Concept: Distance from origin (T4)

Q8. Find the distance of the point (6, 8) from the origin. Which Pythagorean triple is hiding inside this question, and how can the triple help you answer faster next time?

SOLUTION

Step 1: Given: point (6, 8). Needed: distance from O(0, 0).

Step 2: Distance from origin = $\sqrt{(x^2 + y^2)} = \sqrt{(6^2 + 8^2)}$.

Step 3: Compute the squares: 36 + 64 = 100.

Step 4: Take the root: $\sqrt{100} = 10$.

Step 5: The hidden triple is 3-4-5 scaled by 2: 6-8-10. If you spot (6, 8), you can jump straight to 10 without squaring.

Step 6: So the distance is 10 units.

» **Why this step?** Spotting the triple is a safe shortcut only when BOTH numbers of a known triple appear exactly; otherwise use the formula.

Hinglish: *Origin se distance ka short formula: $\sqrt{(x^2 + y^2)}$. 6 aur 8 ko dekhte hi 3-4-5 wala triple yaad karo - dono numbers 2 se multiply ho rahe hain, isliye answer 5 ka 2 guna = 10. Triples ratt lo: 3-4-5, 5-12-13, 8-15-17, 7-24-25. Par dhyan rahe - triple tabhi use karo jab numbers EXACT match karein, warna seedha formula lagao.*

Answer: 10 units; the triple 6-8-10 (which is 2 x (3-4-5)).

Q9 · Basic · Concept: Sign reasoning (T1/T13)

Q9. A point has coordinates (a, -a) where a is a non-zero real number. (i) In which quadrant does it lie when a is positive? (ii) In which quadrant when a is negative? (iii) What happens at a = 0? (iv) What is always true about the distances of this point from the two axes?

SOLUTION

Step 1: Given: the family of points (a, -a). Needed: quadrant analysis by sign of a.

Step 2: Case a positive: x = a is positive, y = -a is negative, so the point is in Quadrant IV.

Step 3: Case a negative: writing a = -m with m positive gives x = -m (negative) and y = m (positive), so the point is in Quadrant II.

Step 4: Case a = 0: the point is (0, 0), the origin itself - on both axes, in no quadrant.

Step 5: For every a, the distance from the y-axis is |a| and the distance from the x-axis is |-a| = |a|. These are always equal.

Step 6: So every point of this family stays exactly equally far from both axes - the family slides along the two lines y = -x.

Hinglish: *Ye question sign-logic ka gym hai. a positive toh x positive, y negative = Quadrant IV. a negative karo toh dono ke sign flip ho jaate hain: x negative, y positive = Quadrant II. a = 0 par origin. Aur sundar observation: is point ki dono axis se doori hamesha barabar hai |a|, kyunki x aur y ka size same hai, sirf sign ulta hai. Aage Ch 2 mein y = -x line padhoge, ye usi ka preview hai.*

Answer: (i) Quadrant IV when a > 0; (ii) Quadrant II when a < 0; (iii) the origin when a = 0; (iv) distances from both axes are always equal (each equals |a|).

Q10 · Basic · Concept: Axis membership (T1/T13)

Q10. From the list (2, 0), (0, 2), (-5, 0), (0, -5), (7, 0), (0, 7), (-3, -3), pick out (i) all points on the x-axis, (ii) all points on the y-axis, (iii) any point equidistant from both axes, and (iv) state how many of the listed points lie in a quadrant.

SOLUTION

Step 1: Given: eight points. Needed: classify by axis membership and symmetry.

Step 2: x-axis means y = 0: the points (2, 0), (-5, 0), (7, 0) qualify.

Step 3: y-axis means x = 0: the points (0, 2), (0, -5), (0, 7) qualify.

Step 4: Equidistant from both axes means |x| = |y|. Only (-3, -3) satisfies this (both distances 3).

Step 5: Six points lie on the axes, and exactly one point, (-3, -3), lies strictly inside a quadrant (Quadrant III).

Hinglish: *Simple filter lagao: y = 0 hai toh x-axis ka banda, x = 0 hai toh y-axis ka. Dono conditions fail ho jaayein tabhi point kisi quadrant ke andar hai. Yahan sirf (-3, -3) quadrant mein girta hai. Aur |x| = |y| wala idea yaad rakho - dono axis se barabar doori ka matlab. Ye chhota sa observation aage bahut bade questions kholta hai.*

Answer: (i) (2, 0), (-5, 0), (7, 0) (ii) (0, 2), (0, -5), (0, 7) (iii) (-3, -3) (iv) exactly 1 point lies in a quadrant (Q III).

Q11 · Moderate · Concept: Distance formula (T4)

Q11. Find the distance between A(1, 2) and B(4, 6) using (i) the helper-triangle method and (ii) the direct formula. Confirm both give the same value.

SOLUTION

Step 1: Given: A(1, 2), B(4, 6). Needed: AB by two methods.

Step 2: Helper triangle: the horizontal leg is |4 - 1| = 3 and the vertical leg is |6 - 2| = 4.

Step 3: By Pythagoras, AB = $\sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4: Direct formula: AB = $\sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(3^2 + 4^2)} = 5$.

Step 5: Both methods agree: AB = 5 units. The helper triangle simply explains where the formula comes from.

» **Why this step?** Showing both methods builds the habit of verifying with geometry - on hard questions this catches sign slips instantly.

Hinglish: *Yahi chapter ka hero question hai. Helper triangle mein humne sirf differences liye - 3 aur 4 - aur wo 3-4-5 triple nikla, isliye answer 5. Formula bhi wahi karta hai, bas ek line mein. Dono ka jawab same aana hi chahiye kyunki formula Pythagoras ka hi shortcut hai. Aage se seedha formula use karo, par agar kabhi doubt ho, helper triangle bana lo.*

Answer: AB = 5 units by both methods.

Q12 · Moderate · Concept: Distance formula with negatives (T4)

Q12. Find the distance between P(-2, 3) and Q(4, -5). Then find the distance of the midpoint of PQ from the origin.

SOLUTION

Step 1: Given: P(-2, 3), Q(4, -5). Needed: PQ, then distance of midpoint M from O.

Step 2: Differences: x-difference = 4 - (-2) = 6; y-difference = -5 - 3 = -8.

Step 3: PQ = $\sqrt{(6^2 + (-8)^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$. (The minus vanished on squaring, as promised.)

Step 4: Midpoint M = ((-2 + 4)/2, (3 + (-5))/2) = (2/2, -2/2) = (1, -1).

Step 5: Distance of M from origin = $\sqrt{(1^2 + (-1)^2)} = \sqrt{2}$.

Step 6: So PQ = 10 units and OM = $\sqrt{2}$ units.

» **Why this step?** Squaring (-8) gives 64 - the sign never survives a square, so never 'pre-correct' it to 8.

Hinglish: *Negatives ke saath formula lagane mein sabse badi galti: log -8 ko 8 karke beech mein hi root le lete hain. Aisa kuch nahi karna - square laga do, sign khud khatam ho jayega. $6^2 + (-8)^2 = 36 + 64 = 100$, root 10. Midpoint = average of coordinates = (1, −1). Aur (1, −1) ka origin se distance $\sqrt{2}$. Chhota point: $\sqrt{2}$ ko decimal mein mat likho jab tak manga na ho.*

Answer: PQ = 10 units; OM = $\sqrt{2}$ units.

Q13 · Moderate · Concept: Collinearity test (T9)

Q13. Without plotting, check whether M(0, 0), A(3, 4) and G(6, 8) lie on the same straight line. State the reasoning principle you used.

SOLUTION

Step 1: Given: three points. Needed: collinearity verdict without plotting.

Step 2: Compute OM = $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.

Step 3: Compute AG = $\sqrt{((6 - 3)^2 + (8 - 4)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4: Compute MG = $\sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$.

Step 5: Test: the largest distance (10) equals the sum of the other two (5 + 5 = 10).

Step 6: Since MG = OM + AG exactly, the point A sits on the straight path from M to G; the three points are collinear.

Step 7: Principle used: on a straight line, the whole equals the sum of the parts (equality case of the triangle inequality). Any detour would make the total strictly larger.

Hinglish: *Ye 'triangle inequality equality' ka test hai - bina graph banaye line check karne ka tareeka. Teen distances nikaalo: 5, 5, 10. Agar sabse bada = baaki dono ka sum, toh teeno ek hi line par hain. Agar sum bada aaye (strictly), toh beech ka point line se hat ke baitha hai. Yahan 5 + 5 = 10 exact match - collinear. Ek baat yaad rakho: values EXACT rakho, round mat karo, warna test galat ho sakta hai.*

Answer: Yes - OM + AG = MG (5 + 5 = 10), so M, A, G are collinear.

Q14 · Moderate · Concept: Reflection chain (T7)

Q14. The point P(4, -3) is reflected in the x-axis to get P_1, and then P_1 is reflected in the y-axis to get P_2. Write the coordinates of P_1 and P_2, and find the length PP_2.

SOLUTION

Step 1: Given: P(4, -3) and a chain of two reflections. Needed: images and PP_2.

Step 2: Reflection in the x-axis keeps x and flips the sign of y: $P_1 = (4, 3)$.

Step 3: Reflection in the y-axis keeps y and flips the sign of x: $P_2 = (-4, 3)$.

Step 4: PP_2 joins (4, -3) to (-4, 3): x-difference = -4 - 4 = -8, y-difference = 3 - (-3) = 6.

Step 5: $PP_2 = \sqrt{((-8)^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 6: Check with preserved distances: $PP_1 = |3 - (-3)| = 6$ (vertical segment), $P_1P_2 = |(-4) - 4| = 8$ (horizontal segment), and indeed $6^2 + 8^2 = 10^2$, so the numbers are internally consistent.

Hinglish: *Reflection ke do simple rules: x-axis mein ungli x-axis par - sirf y ka sign palto. y-axis mein - sirf x ka sign palto. Chain mein dono lagane se (4, -3) pehle (4, 3) bana, phir (-4, 3). Ab distance: 8 aur 6 ka 8-6-10 triple! Aur verification sundar hai: $PP_1 = 6$ vertical, $P_1P_2 = 8$ horizontal, inka Pythagoras bhi 10 hi deta hai. Double confirmation mil gaya.*

Answer: $P_1 = (4, 3)$, $P_2 = (-4, 3)$, $PP_2 = 10$ units.

Q15 · Moderate · Concept: Triangle classification (T6)

Q15. The vertices of a triangle are A(1, 2), B(5, 2) and C(3, 6). Find all three side lengths and classify the triangle. Is it equilateral? Is it right-angled?

SOLUTION

Step 1: Given: three vertices. Needed: side lengths and classification.

Step 2: AB: same y-coordinate (2), so AB = |5 - 1| = 4.

Step 3: AC = $\sqrt{((3 - 1)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$.

Step 4: BC = $\sqrt{((3 - 5)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$.

Step 5: Two sides equal (AC = BC), one different: the triangle is isosceles, not equilateral.

Step 6: Right-angle check by the converse of Pythagoras: is the square of the longest side equal to the sum of squares of the other two? Longest is $2\sqrt{5}$, its square 20. Others: $4^2 = 16$ and $(2\sqrt{5})^2 = 20$, and 16 + 20 = 36 which is not 20.

Step 7: Not right-angled. In fact $4 + 2\sqrt{5} + 2\sqrt{5}$ is the perimeter = $4 + 4\sqrt{5}$.

Hinglish: *Side nikaalne ke baad classification ka recipe: (1) do barabar? isosceles. (2) teen barabar? equilateral. (3) Pythagoras satisfy? right-angled. Yahan AC aur BC dono $2\sqrt{5}$ nikle - isosceles pakka. Right angle ka test: sabse lambi side ka square = baaki dono ke squares ka sum? 20 vs 36 - fail. Toh na equilateral, na right. Simple hai, bas teeno tests order mein lagao.*

Answer: AB = 4, AC = BC = $2\sqrt{5}$; the triangle is isosceles; it is neither equilateral nor right-angled.

Q16 · Moderate · Concept: Square verification (T6)

Q16. Show that the points (2, 2), (6, 2), (6, 6) and (2, 6) are the corners of a square. Then find its area and the length of its diagonal.

SOLUTION

Step 1: Given: four points in cyclic order. Needed: prove square, then area and diagonal.

Step 2: Bottom side (2, 2) to (6, 2): horizontal, length 4.

Step 3: Right side (6, 2) to (6, 6): vertical, length 4.

Step 4: Top side (6, 6) to (2, 6): horizontal, length 4. Left side (2, 6) to (2, 2): vertical, length 4.

Step 5: All four sides equal to 4, and adjacent sides meet at the corners of an axis-aligned rectangle (horizontal meets vertical), so every corner angle is 90 degrees.

Step 6: Hence the figure is a square of side 4. Area = side^2 = 16 square units.

Step 7: Diagonal d joins (2, 2) to (6, 6): d = $\sqrt{((6 - 2)^2 + (6 - 2)^2)} = \sqrt{32} = 4\sqrt{2}$. This matches the square-diagonal relation d = side x $\sqrt{2}$.

Hinglish: *Square prove karne ka two-step formula: (1) chaaron sides barabar, (2) koi bhi ek corner ka angle 90 (yahan horizontal aur vertical lines mil rahe hain toh automatic 90). Side 4 hai, toh area 16 aur diagonal side x $\sqrt{2} = 4\sqrt{2}$. Distance formula se bhi verify kiya - $\sqrt{32} = 4\sqrt{2}$, dono match. Aisi cross-checking exam mein full marks dilati hai.*

Answer: It is a square of side 4; area = 16 sq units; diagonal = $4\sqrt{2}$ units.

Q17 · Moderate · Concept: Right-angle by converse Pythagoras (T6)

Q17. A(1, 1), B(4, 1) and C(4, 5) are three points. Use the distance formula and the converse of the Baudhayana-Pythagoras Theorem to show that angle B is a right angle, and find the area of triangle ABC.

SOLUTION

Step 1: Given: three vertices. Needed: prove right angle at B, then area.

Step 2: AB: same y = 1, so AB = |4 - 1| = 3.

Step 3: BC: same x = 4, so BC = |5 - 1| = 4.

Step 4: AC = $\sqrt{((4 - 1)^2 + (5 - 1)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 5: Check the converse: AC^2 = 25 and AB^2 + BC^2 = 9 + 16 = 25. They are equal, so the angle opposite AC - which is angle B - is a right angle.

Step 6: Area of a right triangle = (1/2) x leg x leg = (1/2) x 3 x 4 = 6 square units.

Step 7: The hidden triple is again 3-4-5.

Hinglish: *Converse Pythagoras ka use yaad karo: agar $c^2 = a^2 + b^2$, toh c ke saamne wala angle 90 hai. Yahan AC sabse lambi hai (5), aur $3^2 + 4^2 = 5^2$ nikla, toh B par right angle. Area ke liye right triangle ka bonus: do legs ko base-height maan lo, (1/2)(3)(4) = 6. Coordinate geometry mein area aise hi nikalte hain jab triangle axis-aligned legs rakhta hai.*

Answer: Angle B = 90 degrees (since $3^2 + 4^2 = 5^2 = AC^2$); area = 6 square units.

Q18 · Moderate · Concept: Missing value, two roots (T5)

Q18. The distance between the points (k, 3) and (4, 3) is 7. Find all possible values of k. How many points satisfy the condition?

SOLUTION

Step 1: Given: two points sharing y = 3, distance 7. Needed: k.

Step 2: Both points have y-coordinate 3, so the segment is horizontal and distance = |x2 - x1|.

Step 3: So |4 - k| = 7. An absolute-value equation splits into two cases.

Step 4: Case 1: 4 - k = 7, giving k = -3. Case 2: 4 - k = -7, giving k = 11.

Step 5: Verification: from (4, 3) to (-3, 3): |4 - (-3)| = 7. From (4, 3) to (11, 3): |11 - 4| = 7. Both work.

Step 6: So there are TWO points, (-3, 3) and (11, 3), one on each side of (4, 3).

» **Why this step?** The absolute value MUST open into two cases. Dropping one case is the single most common error in 'find k' questions.

Hinglish: *Yahi wo question hai jahan aadhe marks ud jaate hain! |4 - k| = 7 ke DO solutions hote hain: point (4, 3) ke left mein bhi ek aur right mein bhi ek. k = -3 aur k = 11. Agar tumne sirf ek nikala, toh samajh lo absolute value ke do case khulane bhool gaye. Verification bhi kar li - dono se distance 7 hi aati hai.*

Answer: k = -3 or k = 11; there are two such points.

Q19 · Moderate · Concept: Circle membership (T10)

Q19. A circle has centre the origin and radius 5. Classify each point as on, inside, or outside the circle: (3, 4), (4, 2), (-3, 4), (0, 5), (-4, -4).

SOLUTION

Step 1: Given: circle $x^2 + y^2 = 25$. Needed: classify five points.

Step 2: Method: compare $x^2 + y^2$ with $r^2 = 25$. Equal means on the circle; smaller means inside; larger means outside.

Step 3: (3, 4): 9 + 16 = 25, exactly equal, so ON the circle.

Step 4: (4, 2): 16 + 4 = 20, which is less than 25, so INSIDE (distance $\sqrt{20}$ slightly under 5).

Step 5: (-3, 4): 9 + 16 = 25, so ON the circle (signs never matter after squaring).

Step 6: (0, 5): 0 + 25 = 25, so ON the circle.

Step 7: (-4, -4): 16 + 16 = 32, greater than 25, so OUTSIDE (distance $\sqrt{32} = 4\sqrt{2}$ approx 5.66).

» **Why this step?** Comparing squares instead of square roots avoids every rounding issue - 20 vs 25 is cleaner than 4.47 vs 5.

Hinglish: *Circle membership ka super-fast test: $x^2 + y^2$ ko r^2 se compare karo. Barabar = line par, kam = andar, zyada = bahar. Root nikalne ki zaroorat hi nahi. (-3, 4) dekho - minus hone ke bawajood square ke baad 25 hi aaya, toh circle par hi hai. (-4, -4) dono square 16+16 = 32 > 25, bahar. Ye technique time bhi bachati hai aur rounding errors se bhi bachati hai.*

Answer: On the circle: (3, 4), (-3, 4), (0, 5). Inside: (4, 2). Outside: (-4, -4).

Q20 · Moderate · Concept: Missing y-value on a circle (T5/T10)

Q20. The point (3, y) lies at distance 5 from the origin. Find all possible values of y. Where do the two possible points sit relative to each other?

SOLUTION

Step 1: Given: distance from O equals 5. Needed: y.

Step 2: Distance from origin: $\sqrt{(3^2 + y^2)} = 5$.

Step 3: Square both sides (both sides are non-negative, so squaring is safe): $9 + y^2 = 25$.

Step 4: So $y^2 = 16$, and taking square roots gives y = 4 or y = -4.

Step 5: Verification: (3, 4) gives $\sqrt{(9 + 16)} = 5$, and (3, -4) also gives 5. Both are genuine solutions.

Step 6: The two points (3, 4) and (3, -4) are reflections of each other in the x-axis - same x, opposite y - and both lie on the circle of radius 5.

Hinglish: *Yahan bhi do answers aayenge, aur wajah samjho: circle ke ek vertical line par do points hote hain - ek upar, ek neeche. y = 4 aur y = -4. Dono verify kiye, dono sahi. Aur ye dono points x-axis ke reflection partners hain. Ye pattern (quadratic se do roots) Chapter 1 ka signature hai - jab bhi circle ya distance se equation bane, do*

answers ki ummeed rakho.

Answer: y = 4 or y = -4; the two points are mirror images in the x-axis, both on the circle of radius 5.

Q21 · Difficult · Concept: Perimeter + classification (T6/T12)

Q21. Find the perimeter of the triangle with vertices A(2, -2), B(-2, 1) and C(5, 2). Along the way, show that the triangle is isosceles.

SOLUTION

Step 1: Given: three vertices. Needed: all sides, isosceles check, perimeter.

Step 2: AB: differences are (-2 - 2) = -4 and (1 - (-2)) = 3. AB = √(16 + 9) = √25 = 5.

Step 3: BC: differences are (5 - (-2)) = 7 and (2 - 1) = 1. BC = √(49 + 1) = √50 = 5√2.

Step 4: CA: differences are (2 - 5) = -3 and (-2 - 2) = -4. CA = √(9 + 16) = √25 = 5.

Step 5: AB = CA = 5, so the triangle is isosceles with apex at A.

Step 6: Perimeter = AB + BC + CA = 5 + 5√2 + 5 = 10 + 5√2 units (approximately 17.07).

Step 7: Keep the exact form 10 + 5√2 in the answer; convert to decimals only if the question asks.

Hinglish: *Do sides 5 nikli aur ek 5√2 - isosceles confirm. Perimeter mein exact value rakho: 10 + 5√2. Exam mein log √50 ko 7.07 leke ghumaa dete hain aur final answer gandaa ho jata hai. Rule yaad rakho: jab tak decimal manga nahi, root form hi rakho. Aur ek observation: (4, 3) wala 3-4-5 triple do baar mila - coordinate questions mein triples baar-baar ghuumte hain, unhe pakadna speed deta hai.*

Answer: Perimeter = 10 + 5√2 units; the triangle is isosceles (AB = CA = 5).

Q22 · Difficult · Concept: Reverse midpoint (T8)

Q22. M(-1, 3) is the midpoint of the segment joining A(2, -4) and B(x, y). Find B, and verify your answer by checking that MA = MB.

SOLUTION

Step 1: Given: midpoint and one endpoint. Needed: the other endpoint.

Step 2: Midpoint equations: (-1) = (2 + x)/2 and 3 = (-4 + y)/2.

Step 3: Solve the first: multiply by 2: -2 = 2 + x, so x = -4.

Step 4: Solve the second: 6 = -4 + y, so y = 10.

Step 5: So B = (-4, 10).

Step 6: Verify MA: differences (2 - (-1)) = 3 and (-4 - 3) = -7, MA = √(9 + 49) = √58.

Step 7: Verify MB: differences (-4 - (-1)) = -3 and (10 - 3) = 7, MB = √(9 + 49) = √58. Equal, so B is correct.

» Why this step? Reverse midpoint = undo the average: multiply by 2, then subtract the known endpoint. Verification with equal distances is cheap and catches slips.

Hinglish: *Midpoint wala equation ulta chalana hai: average diya hai, original nikalna hai. 2 se multiply karo, phir known endpoint minus karo. x ke liye: (-1)(2) - 2 = -4. y ke liye: (3)(2) - (-4) = 10. Verification zaroor karo - MA aur MB dono √58 aaye, matlab M sach mein beech mein baitha hai. Bina verification ke answer likhna adhoora hai.*

Answer: B = (-4, 10), verified since MA = MB = √58.

Q23 · Difficult · Concept: Parallelogram check (T6)

Q23. Show that the points A(0, 0), B(4, 0), C(6, 3), D(2, 3) form a parallelogram, and then show that it is NOT a rectangle.

SOLUTION

Step 1: Given: four points in order. Needed: parallelogram yes, rectangle no.

Step 2: AB: from (0, 0) to (4, 0): horizontal, length 4.

Step 3: CD: from (6, 3) to (2, 3): horizontal, length |6 - 2| = 4. So AB = CD and both are horizontal - parallel and equal.

Step 4: BC: from (4, 0) to (6, 3): $\sqrt{(2^2 + 3^2)} = \sqrt{13}$. DA: from (2, 3) to (0, 0): $\sqrt{(2^2 + 3^2)} = \sqrt{13}$. So BC = DA, both with the same slant.

Step 5: Both pairs of opposite sides are equal (and clearly parallel: horizontal pairs, and BC, DA share the same step pattern 2-across, 3-up). So ABCD is a parallelogram.

Step 6: Rectangle test: a parallelogram is a rectangle exactly when its diagonals are equal.

Step 7: AC = $\sqrt{(6^2 + 3^2)} = \sqrt{45} = 3\sqrt{5}$. BD = $\sqrt{((2 - 4)^2 + (3 - 0)^2)} = \sqrt{(4 + 9)} = \sqrt{13}$.

Step 8: Since $3\sqrt{5}$ is not equal to $\sqrt{13}$, the diagonals are unequal, so the parallelogram is not a rectangle.

» **Why this step?** The diagonal test is the cleanest rectangle check using ONLY distance - no angle measurement needed.

Hinglish: *Parallelogram proof ke liye opposite sides barabar kafi hain (parallel toh step pattern se dikh hi raha hai). Rectangle check ka shortcut yaad rakho: parallelogram mein diagonals barabar hona = rectangle. Yahan AC = 3√5 aur BD = √13 - barabar nahi, toh rectangle nahi. Ye diagonal-test chapter ke andar hi aata hai kyunki wo sirf distance formula use karta hai. Sketch banaoge toh slanted box dikhega - jaise side se dhakela hua rectangle.*

Answer: AB = CD = 4 and BC = DA = √13, so ABCD is a parallelogram; its diagonals $3\sqrt{5}$ and √13 are unequal, so it is not a rectangle.

Q24 · Difficult · Concept: Equidistant point on an axis (T5/T8)

Q24. Find the point on the y-axis that is equidistant from P(-2, 0) and Q(6, 4). Verify your answer.

SOLUTION

Step 1: Given: unknown point R on the y-axis. Needed: R equidistant from P and Q.

Step 2: On the y-axis the point has the form R(0, y) - fix the zero first.

Step 3: $RP^2 = (0 - (-2))^2 + (y - 0)^2 = 4 + y^2$.

Step 4: $RQ^2 = (0 - 6)^2 + (y - 4)^2 = 36 + (y - 4)^2$.

Step 5: Set them equal: $4 + y^2 = 36 + y^2 - 8y + 16$.

Step 6: Cancel y^2 from both sides (this is why the equation stays linear and easy): $4 = 52 - 8y$.

Step 7: So $8y = 48$, giving $y = 6$. Hence R = (0, 6).

Step 8: Verify: RP = $\sqrt{(4 + 36)} = \sqrt{40} = 2\sqrt{10}$. RQ = $\sqrt{(36 + (6 - 4)^2)} = \sqrt{(36 + 4)} = \sqrt{40} = 2\sqrt{10}$. Equal, so R is correct.

» **Why this step?** Comparing squared distances avoids square roots entirely; and y^2 cancels, leaving a linear equation.

Hinglish: *Y-axis par point = (0, y) - pehle yeh lock karo. Phir dono distances ke SQUARES barabar rakho, root ka jhanjhat khatam. Sabse sundar cheez: y^2 dono taraf cancel ho gaya aur seedha linear equation mil gaya, y = 6. Verify karo: dono distances 2√10. Aisa equidistant point technically perpendicular bisector par baithta hai - yahan*

humne uska sirf y-axis wala intersection nikala. 2√10 ko 6.32 likhne ki zaroorat nahi.

Answer: R = (0, 6), verified since RP = RQ = 2√10.

Q25 · Difficult · Concept: Right isosceles triangle (T6/T14)

Q25. A right-angled isosceles triangle has its right angle at the origin and its two equal legs of length 6 lying along the positive x-axis and positive y-axis. Write the coordinates of all three vertices, find the hypotenuse, and find the area. Generalise: if the legs are a, what is the hypotenuse?

SOLUTION

Step 1: Given: right angle at O, legs 6 along positive axes. Needed: vertices, hypotenuse, area, general rule.

Step 2: One vertex is O(0, 0). Walking 6 along the positive x-axis gives (6, 0); walking 6 up the positive y-axis gives (0, 6).

Step 3: Hypotenuse joins (6, 0) to (0, 6): length = $\sqrt{(6^2 + 6^2)} = \sqrt{72} = 6\sqrt{2}$.

Step 4: Area = (1/2) x base x height = (1/2) x 6 x 6 = 18 square units.

Step 5: General case: legs a and a give hypotenuse $\sqrt{(a^2 + a^2)} = \sqrt{(2a^2)} = a\sqrt{2}$.

Step 6: This is exactly the PDF's special-case observation: putting a = b in the general Pythagoras relation forces hypotenuse = a√2 for isosceles right triangles.

Hinglish: *Right isosceles ka signature result: hypotenuse hamesha leg ka √2 guna hota hai. 6 ka leg toh 6√2. Yaad rakhne ka trick: square ka diagonal bhi side ka √2 guna hota hai - same baat hai, kyunki aadha square hi toh ye triangle hai! Area half-product = 18. Generalisation likhna (a√2) exam mein extra impression banata hai kyunki PDF mein special-case wala box isi baat ko highlight karta hai.*

Answer: Vertices (0, 0), (6, 0), (0, 6); hypotenuse = 6√2; area = 18 sq units; in general the hypotenuse is a√2.

Q26 · Difficult · Concept: City-grid application (T11)

Q26. In a city, streets run in two perpendicular directions, 200 m apart. Intersections are labelled (row, column). An auto rickshaw goes from intersection (1, 1) to intersection (4, 3). (i) What is the shortest distance it can travel moving only along streets? (ii) What is the straight-line (as-the-crow-flies) distance? (iii) How much shorter is the direct route?

SOLUTION

Step 1: Given: grid spacing 200 m, start (1, 1), end (4, 3). Needed: street distance, direct distance, saving.

Step 2: Moving along streets from row 1 to row 4 needs 3 steps; from column 1 to column 3 needs 2 steps.

Step 3: Shortest street path = 3 + 2 = 5 steps = 5 x 200 = 1000 m.

Step 4: Direct distance uses the distance formula with coordinate gap 3 rows and 2 columns: $\sqrt{(3^2 + 2^2)}$ x 200 = √13 x 200 approx 3.606 x 200 approx 721 m.

Step 5: Saving = 1000 - 200√13 approx 1000 - 721 = 279 m (about 28 percent).

Step 6: Note the concept: along-streets distance adds the gaps; direct distance applies Pythagoras to the same gaps.

Hinglish: *City grid hi bada coordinate plane hai - bas unit 200 m hai. Galiyon mein chalna hai toh horizontal + vertical steps JODO (3 + 2 = 5 blocks = 1000 m). Seedha udna hai toh Pythagoras lagao: √(9 + 4) = √13 blocks. Difference dekho - seedha rasta karib 280 m bachata hai. Yahi idea PDF ke athletics track aur map questions mein bhi chalta hai: grid distance vs straight distance.*

Answer: (i) 1000 m (ii) 200√13 approx 721 m (iii) approx 279 m shorter

Q27 · Difficult · Concept: Rhombus from coordinates (T6)

Q27. The points (0, 5), (-3, 0), (0, -5) and (3, 0) are the corners of a quadrilateral taken in order. Show that it is a rhombus and find the lengths of its diagonals and its area.

SOLUTION

Step 1: Given: four symmetric points. Needed: prove rhombus, diagonals, area.

Step 2: Side from (0, 5) to (-3, 0): $\sqrt{(3^2 + 5^2)} = \sqrt{34}$.

Step 3: Side from (-3, 0) to (0, -5): $\sqrt{(3^2 + 5^2)} = \sqrt{34}$. By the same computation each of the remaining two sides is $\sqrt{(3^2 + 5^2)} = \sqrt{34}$.

Step 4: All four sides are equal to $\sqrt{34}$, so the quadrilateral is a rhombus (a parallelogram with equal sides; symmetry about both axes guarantees the parallelogram behaviour, and can be double-checked: opposite sides are translations of each other).

Step 5: Diagonal 1 joins (0, 5) to (0, -5): vertical, length 10.

Step 6: Diagonal 2 joins (-3, 0) to (3, 0): horizontal, length 6.

Step 7: Area of a rhombus = (1/2) x product of diagonals = (1/2) x 10 x 6 = 30 square units.

Hinglish: *Chaaron sides √34 - rhombus pakka. Diagonals yahan axis par hi baithi hain, isliye unki length seedhi mili: 10 aur 6. Rhombus ka area = (1/2) d1 d2 = 30 - ye formula yaad rakho, square/rectangle ke bhai ki tarah. Ye PDF ke Ch 6 se connect hota hai jahan rhombus-area questions hain, par classification toh pure Chapter 1 ka khel hai (distance formula).*

Answer: All sides √34, so it is a rhombus; diagonals 10 and 6; area = 30 square units.

Q28 · Difficult · Concept: Algebraic distance (T4/T14)

Q28. Find the distance between the points (a + b, a - b) and (a - b, a + b), where a and b are real numbers with b not equal to 0. The answer must be in terms of b.

SOLUTION

Step 1: Given: two symbolic points. Needed: distance in terms of b.

Step 2: x-difference: (a + b) - (a - b) = 2b.

Step 3: y-difference: (a - b) - (a + b) = -2b.

Step 4: Distance = $\sqrt{((2b)^2 + (-2b)^2)} = \sqrt{(4b^2 + 4b^2)} = \sqrt{(8b^2)}$.

Step 5: Simplify: $\sqrt{(8b^2)} = \sqrt{8} \times \sqrt{(b^2)} = 2\sqrt{2} \times |b|$.

Step 6: Since b is not zero, for positive b this is $2\sqrt{2}$ b. The cleanest exact answer is $2\sqrt{2}\,|b|$; note that a cancels out completely - the distance does not depend on a at all.

» Why this step? Notice how the a-terms cancel inside both differences. Spotting cancellations before squaring is a mark of an algebra-fluent student.

Hinglish: *Symbolic distance question - numbers ki jagah letters, darne ki zaroorat nahi. Difference lo, a dono jagah cancel ho gaya, bacha 2b aur -2b. Square ke baad $8b^2$, root se $2\sqrt{2}\,|b|$. Bar (absolute value) isliye kyunki distance kabhi negative nahi hoti. Interesting observation: answer a par depend hi nahi karta - dono points a se same doori par slide karte hain, unke beech ka gap sirf b se banta hai.*

Answer: Distance = $\sqrt{(8b^2)} = 2\sqrt{2}\,|b|$ (independent of a).

Q29 · Difficult · Concept: Equidistant locus (T5/T14)

Q29. A point P(x, y) moves so that its distance from A(3, 4) always equals its distance from B(-1, 6). Show that x and y must satisfy the relation y = 2x + 3, and verify that the point (0, 3) obeys it.

SOLUTION

Step 1: Given: PA = PB for a moving point. Needed: the relation between x and y.

Step 2: Write squared distances (roots dropped on both sides - legal since both sides are non-negative).

Step 3: $PA^2 = (x - 3)^2 + (y - 4)^2 = x^2 - 6x + 9 + y^2 - 8y + 16$.

Step 4: $PB^2 = (x + 1)^2 + (y - 6)^2 = x^2 + 2x + 1 + y^2 - 12y + 36$.

Step 5: Set $PA^2 = PB^2$: $x^2 - 6x + 9 + y^2 - 8y + 16 = x^2 + 2x + 1 + y^2 - 12y + 36$.

Step 6: Cancel x^2 and y^2 from both sides: $-6x - 8y + 25 = 2x - 12y + 37$.

Step 7: Move everything to one side: $-8x + 4y - 12 = 0$, that is $4y = 8x + 12$, giving $y = 2x + 3$.

Step 8: Verify (0, 3): $PA = \sqrt{((0 - 3)^2 + (3 - 4)^2)} = \sqrt{(9 + 1)} = \sqrt{10}$. $PB = \sqrt{((0 + 1)^2 + (3 - 6)^2)} = \sqrt{(1 + 9)} = \sqrt{10}$. Equal - the point obeys the condition.

Hinglish: *Ye Chapter 1 ka sabse powerful idea hai: distance ki equality ek RELATION bana deti hai. Square karke x^2 aur y^2 cancel kar do - jo bacha wo straight line hai, $y = 2x + 3$. Is line ke har point dono A aur B se barabar door hai (isko perpendicular bisector kehte hain - Class 11 mein formally aayega). Verification: (0, 3) se dono distances $\sqrt{10}$. Jab bhi 'equidistant from two points' dikhe, yahi square-and-cancel recipe chalao.*

Answer: PA = PB forces y = 2x + 3; verified at (0, 3) where PA = PB = $\sqrt{10}$.

Q30 · Difficult · Concept: Circle membership with bigger radius (T10)

Q30. Consider the circle with centre the origin and radius 13. (i) Verify that (5, 12), (12, 5), (-12, 5) and (0, 13) all lie on the circle. (ii) Determine whether (6, 12) and (4, 12) lie inside, on, or outside it.

SOLUTION

Step 1: Given: circle $x^2 + y^2 = 169$. Needed: classify six points.

Step 2: (5, 12): 25 + 144 = 169, so on the circle (the 5-12-13 triple).

Step 3: (12, 5): 144 + 25 = 169, on the circle (same triple, swapped).

Step 4: (-12, 5): 144 + 25 = 169, on the circle (signs vanish on squaring).

Step 5: (0, 13): 0 + 169 = 169, on the circle (topmost point).

Step 6: (6, 12): 36 + 144 = 180 which is greater than 169, so OUTSIDE the circle.

Step 7: (4, 12): 16 + 144 = 160 which is less than 169, so INSIDE the circle.

» Why this step? The 5-12-13 triple is the second most useful triple after 3-4-5; radius 13 questions are built around it.

Hinglish: *Radius 13 = 5-12-13 triple ka ghar. Chaaron points check karo - sab 169 de rahe hain, sab circle par. Phir (6, 12) aur (4, 12) ka fun part: 180 > 168... nahi, 180 > 169 toh bahar; 160 < 169 toh andar. Dekho kamaal: (6, 12) aur (4, 12) dono lagbhag same jagah hain, par ek bahar aur ek andar! Circle ki seedhi borderline - yahi precision Chapter 1 sikhata hai.*

Answer: All four triple points lie on the circle; (6, 12) is outside; (4, 12) is inside.

Q31 · High · Concept: Trisection points (T8)

Q31. P and Q are the points of trisection of the segment AB, with P closer to A and Q closer to B, where A(4, 7) and B(16, -2). Find P and Q. Then verify that the midpoint of PQ coincides with the midpoint of AB.

SOLUTION

Step 1: Given: A(4, 7), B(16, -2), trisection needed. Needed: P, Q, and the midpoint check.

Step 2: Step size: one third of the way from A to B means adding one third of each gap. Gap in x: 16 - 4 = 12; gap in y: -2 - 7 = -9.

Step 3: One third of the gaps: 12/3 = 4 and -9/3 = -3.

Step 4: P (closer to A) = A + one step = (4 + 4, 7 - 3) = (8, 4).

Step 5: Q = A + two steps = (4 + 8, 7 - 6) = (12, 1). Equivalently Q = B - one step = (16 - 4, -2 + 3) = (12, 1). Both give the same.

Step 6: Midpoint of P and Q = ((8 + 12)/2, (4 + 1)/2) = (10, 2.5).

Step 7: Midpoint of A and B = ((4 + 16)/2, (7 - 2)/2) = (10, 2.5). They coincide - the check passes.

» Why this step? The middle of the two trisectors is always the midpoint of the whole segment - a beautiful structural fact worth remembering for instant verification.

Hinglish: *Trisection matlab segment ko 3 barabar tukdon mein kaatna. Fundaa simple: total gap ko 3 se divide karo, wo ek 'step' hai. P = A + ek step, Q = A + do step. Yahan steps (4, -3) the. Aur jo self-check diya na - PQ ka midpoint hamesha AB ke midpoint ke barabar hota hai - ye free verification hai, exam mein zaroor lagao. (10, 2.5) = (10, 2.5), perfect match.*

Answer: P(8, 4), Q(12, 1); both midpoints equal (10, 2.5), verification passes.

Q32 · High · Concept: Missing rectangle vertex (T6/T8)

Q32. Three vertices of a rectangle are A(1, 2), B(5, 2) and C(5, 7). Find the fourth vertex D, and verify the rectangle property that its diagonals are equal.

SOLUTION

Step 1: Given: three consecutive vertices. Needed: D and the diagonal check.

Step 2: AB is horizontal (y = 2 for both) with length 4; BC is vertical (x = 5 for both) with length 5.

Step 3: In a rectangle the fourth side DA must be vertical (above A) and CD horizontal - so D must sit directly above A at the same height as C.

Step 4: Directly above A(1, 2) means x = 1; at the same height as C means y = 7. So D = (1, 7).

Step 5: Verify with side lengths: DA from (1, 7) to (1, 2) has length 5, CD from (5, 7) to (1, 7) has length 4 - matching the opposite sides.

Step 6: Diagonal AC = $\sqrt{((5 - 1)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$.

Step 7: Diagonal BD = $\sqrt{((1 - 5)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$. Equal, so the rectangle property holds.

Hinglish: *Rectangle ka chautha vertex pattern se milta hai: A ke seedha upar, C ke level par. x = 1 (A jaisa), y = 7 (C jaisa). Phir property check - diagonals barabar? Dono √41. Yahan √41 ko simplify nahi kar sakte (41 prime hai), toh root form hi final hai. Notice karo: har rectangle mein diagonals barabar hote hain - ye test humne parallelogram question (Q23) mein bhi use kiya tha, wahan fail hua tha, yahan pass.*

Answer: D = (1, 7); diagonals AC = BD = √41, confirming the rectangle.

Q33 · High · Concept: Area of right triangle from coordinates (T12)

Q33. A triangle has vertices O(0, 0), A(8, 0) and B(0, 6). Show that it is right-angled, find its hypotenuse, and compute its area and perimeter.

SOLUTION

Step 1: Given: three vertices, two of them on the axes. Needed: right angle, hypotenuse, area, perimeter.

Step 2: OA lies along the x-axis: length 8. OB lies along the y-axis: length 6.

Step 3: The x-axis and y-axis are perpendicular, and OA, OB lie along them, so the angle at O is 90 degrees.

Step 4: Hypotenuse AB = $\sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$ (the 6-8-10 triple).

Step 5: Area = (1/2) x OA x OB = (1/2) x 8 x 6 = 24 square units.

Step 6: Perimeter = 8 + 6 + 10 = 24 units. A charming coincidence: area 24 and perimeter 24.

Hinglish: *Jab triangle ke do sides axes par baithe hon, right angle free mein milta hai - axis parallel hi perpendicular hote hain. Phir 8-6-10 triple se hypotenuse 10. Area = 24, perimeter bhi 24 - dono number same! Aisi coincidences questions mein jaan-bujh ke rakhi jaati hain, toh apne practice questions mein bhi aisi numbers chuno toh checking easy hoti hai.*

Answer: Right angle at O; hypotenuse AB = 10; area = 24 sq units; perimeter = 24 units.

Q34 · High · Concept: Square from its diagonal (T14)

Q34. Two opposite corners of a square are P(-3, 2) and R(5, -4). Find the length of the diagonal, the side of the square, its area, and the coordinates of the other two corners.

SOLUTION

Step 1: Given: diagonal endpoints of a square. Needed: diagonal, side, area, remaining vertices.

Step 2: Diagonal PR = $\sqrt{((5 - (-3))^2 + (-4 - 2)^2)} = \sqrt{(8^2 + (-6)^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 3: For a square of side s, the diagonal is $s\sqrt{2}$, so $s = 10/\sqrt{2} = 5\sqrt{2}$.

Step 4: Area = $s^2 = (5\sqrt{2})^2 = 50$ square units. (Shortcut: area = diagonal^2/2 = 100/2 = 50.)

Step 5: Centre of the square = midpoint of PR = ((-3 + 5)/2, (2 - 4)/2) = (1, -1).

Step 6: Half-diagonal vector from centre to R is (5 - 1, -4 - (-1)) = (4, -3). To build the other diagonal, swap the components and flip one sign: (3, 4).

Step 7: Other corners: centre + (3, 4) = (4, 3) and centre - (3, 4) = (-2, -5).

Step 8: Verify: Q(4, 3) to P(-3, 2): $\sqrt{(7^2 + 1)} = \sqrt{50} = 5\sqrt{2}$ (a side). Q to R(5, -4): $\sqrt{(1 + 49)} = \sqrt{50}$ (a side). Q to S(-2, -5): $\sqrt{(36 + 64)} = 10$ (the other diagonal). All consistent.

» Why this step? The swap-and-flip move rotates the half-diagonal by 90 degrees - it is just Pythagoras thinking in vector form, and it is fully verifiable with the distance formula.

Hinglish: *Square ka diagonal diya hai toh side = diagonal/$\sqrt{2}$ = $5\sqrt{2}$, area = 50. Baaki do corners nikalne ka tareeka: pehle centre (midpoint) lo - (1, -1). Half-diagonal (4, -3) hai. Uske components SWAP karo aur ek sign flip karo: (3, 4) ban jayega - ye 90 degree ghuma hua half-diagonal hai. Centre mein plus-minus lagao: (4, 3) aur (-2, -5). Phir verification: sab sides $5\sqrt{2}$, doosra diagonal 10. Ye wala question HOTS-zone ka gateway hai - rotation ka idea samajh gaye toh aage Olympiad-level coordinate proofs easy lagenge.*

Answer: Diagonal = 10; side = $5\sqrt{2}$; area = 50 sq units; other corners Q(4, 3) and S(-2, -5).

Q35 · High · Concept: Collinearity - negative coordinates (T9)

Q35. Check, without plotting, whether the points (-2, -1), (0, 0) and (4, 3) lie on one straight line. Show your exact arithmetic.

SOLUTION

Step 1: Given: three points with negative coordinates. Needed: collinearity verdict with exact values.

Step 2: d1 = distance from (-2, -1) to (0, 0) = $\sqrt{(2^2 + 1^2)} = \sqrt{5}$.

Step 3: d2 = distance from (0, 0) to (4, 3) = $\sqrt{(16 + 9)} = \sqrt{25} = 5$.

Step 4: d3 = distance from (-2, -1) to (4, 3) = $\sqrt{(6^2 + 4^2)} = \sqrt{(36 + 16)} = \sqrt{52} = 2\sqrt{13}$.

Step 5: Collinearity requires the largest to equal the sum of the other two: is $2\sqrt{13} = \sqrt{5} + 5$?

Step 6: Compare squares of both sides: left squared is 52. Right squared is $(\sqrt{5} + 5)^2 = 5 + 25 + 10\sqrt{5} = 30 + 10\sqrt{5}$ approx $30 + 22.36 = 52.36$.

Step 7: Since 52 differs from $30 + 10\sqrt{5}$ (equivalently $\sqrt{5} + 5$ approx 7.236 while $2\sqrt{13}$ approx 7.211), the equality FAILS.

Step 8: So the three points are NOT collinear - the middle point sits slightly off the line through the outer two.

» **Why this step?** When radicals are involved, compare by squaring (both sides positive) rather than by decimals - decimals can mislead near-miss cases.

Hinglish: *Negative coordinates aane par bhi process same hai. Yahan twist ye hai ki distances roots mein hain - √5, 5, 2√13. Comparison exact karo: dono side ko square karo. Left 52, right 30 + 10√5 approx 52.36. Thoda sa difference hai - matlab NOT collinear. Ye question isliye rakha hai ki tum consider karo ki values kabhi approximate judge na karo. Agar maine jaldi mein 7.24 vs 7.21 dekh ke 'barabar hai' bol diya hota, toh galat ho jaata!*

Answer: Not collinear: $\sqrt{5} + 5$ is not equal to $2\sqrt{13}$ (52 is not equal to $30 + 10\sqrt{5}$).

Q36 · High · Concept: Applied screen/circle problem (T11/T10)

Q36. A computer screen is a rectangle 800 pixels wide and 600 pixels high, with the origin at the bottom-left corner. A circular icon of radius 80 has centre A(100, 150); another of radius 100 has centre B(250, 230). (i) Does any part of either icon stick out of the screen? (ii) Do the two icons overlap each other?

SOLUTION

Step 1: Given: screen 800 x 600, two circles with centres and radii. Needed: boundary check and overlap check.

Step 2: Icon A: leftmost x = 100 - 80 = 20 (inside, positive), rightmost x = 100 + 80 = 180 (inside, below 800), lowest y = 150 - 80 = 70 (inside), highest y = 150 + 80 = 230 (inside, below 600). Icon A is fully on the screen.

Step 3: Icon B: leftmost x = 250 - 100 = 150, rightmost = 250 + 100 = 350, lowest y = 230 - 100 = 130, highest y = 330. All within 0 to 800 and 0 to 600. Fully on screen too.

Step 4: Overlap test for two circles: compare the centre distance d with r1 + r2 and |r1 - r2|.

Step 5: Centre distance AB = $\sqrt{((250 - 100)^2 + (230 - 150)^2)} = \sqrt{(150^2 + 80^2)} = \sqrt{(22500 + 6400)} = \sqrt{28900}$ = 170 (the 8-15-17 triple scaled by 10).

Step 6: Now r1 + r2 = 180 and |r1 - r2| = 20. Since 20 is less than 170 and 170 is less than 180, the circles intersect in two points - they overlap partially.

Step 7: So: no icon leaves the screen, and the icons DO overlap.

Hinglish: *Screen check: har icon ka extreme left/right/top/bottom compute karo aur 0-800, 0-600 range se compare karo. Dono andar hain. Overlap ka test yaad rakho: centres ki doori vs radii ka sum. Distance 170 nikla (8-15-17 triple x 10!). 170 < 180 isliye overlap hota hai, aur 170 > 20 isliye poora ek doosre ke andar nahi hai - partial overlap. Ye 2-line test high-school geometry ka swiss knife hai.*

Answer: (i) Both icons lie fully inside the screen. (ii) Yes, the icons overlap: AB = 170 lies strictly between |100 - 80| = 20 and 80 + 100 = 180.

Q37 · High · Concept: Locus: equidistant from both axes (T14)

Q37. Describe the set of all points that are equidistant from the x-axis and the y-axis. If such a point must also lie at distance 5 from the origin, list ALL points that satisfy both conditions (exact values).

SOLUTION

Step 1: Given: equidistant from both axes, then distance-5 filter. Needed: locus, then exact points.

Step 2: Distance from the x-axis is |y|; distance from the y-axis is |x|. Equality |x| = |y| means y = x or y = -x.

Step 3: So the locus is the pair of lines y = x and y = -x (the two diagonals through the origin).

Step 4: Now add: $\sqrt{(x^2 + y^2)} = 5$. On the line y = x this becomes $\sqrt{(2x^2)} = \sqrt{2}\,|x| = 5$, so $|x| = 5/\sqrt{2} = (5\sqrt{2})/2$.

Step 5: That gives two points: $((5\sqrt{2})/2, (5\sqrt{2})/2)$ and $(-(5\sqrt{2})/2, -(5\sqrt{2})/2)$.

Step 6: On the line y = -x the same computation gives $((5\sqrt{2})/2, -(5\sqrt{2})/2)$ and $(-(5\sqrt{2})/2, (5\sqrt{2})/2)$.

Step 7: Total: four points, one in each quadrant, all at distance 5 from O and equidistant from both axes. Verify one: $((5\sqrt{2})/2)^2 + ((5\sqrt{2})/2)^2 = 25/2 + 25/2 = 25$, distance 5. Correct.

Hinglish: *Dono axis se barabar door = |x| = |y| = do diagonal lines y = x aur y = -x. Ab distance 5 bhi chahiye: $\sqrt{(x^2 + y^2)} = 5$. y = x rakhne par $2x^2 = 25$, $x = 5/\sqrt{2} = 5\sqrt{2}/2$. Rationalise kar dena toh aur saaf dikhta hai. Chaar points bane - har quadrant mein ek, sab circle (radius 5) par aur dono diagonals par. Locus thinking + circle membership ka combo - yahi HOTS ka taste hai.*

Answer: Locus: the two lines y = x and y = -x. The four points are (plus/minus $5\sqrt{2}/2$, plus/minus $5\sqrt{2}/2$) with matching signs on y = x and opposite signs on y = -x.

Q38 · High · Concept: Triple spotting at scale (T4/T12)

Q38. A triangle has vertices (0, 0), (5, 0) and (0, 12). Identify the right angle, find all three sides, and compute the area and the perimeter. Then state the general rule for a right triangle with legs a and b.

SOLUTION

Step 1: Given: vertices on both axes. Needed: sides, area, perimeter, general rule.

Step 2: The side from (0, 0) to (5, 0) lies on the x-axis (length 5); the side from (0, 0) to (0, 12) lies on the y-axis (length 12).

Step 3: These two axis lines are perpendicular, so the right angle is at the origin.

Step 4: Hypotenuse = $\sqrt{(5^2 + 12^2)} = \sqrt{(25 + 144)} = \sqrt{169} = 13$ (the 5-12-13 triple).

Step 5: Area = (1/2) x 5 x 12 = 30 square units.

Step 6: Perimeter = 5 + 12 + 13 = 30 units. Again area equals perimeter - these triple-based triangles love that coincidence.

Step 7: General rule: for a right triangle with legs a and b, hypotenuse = $\sqrt{(a^2 + b^2)}$, area = ab/2.

Hinglish: *5-12-13 triple ka full Dabangg avatar: area bhi 30, perimeter bhi 30. Chhota triangle 3-4-5 ka bhi area 6 aur perimeter 12 tha - wahan match nahi tha, par 5-12-13 aur 6-8-10 mein area = perimeter nikalta hai (8-6-10: area 24, perimeter 24). Aisi patterns notice karna maths ko interesting banata hai. General rule likhna zaroori hai - examiner ko dikhta hai ki tum case se general tak soch rahe ho.*

Answer: Right angle at the origin; sides 5, 12, 13; area = 30 sq units; perimeter = 30 units; general: hypotenuse $\sqrt{(a^2 + b^2)}$, area ab/2.

Q39 · High · Concept: Rectangle verification kit (T6)

Q39. K(1, 1), L(5, 1), M(5, 4) and N(1, 4) are joined in order. Using distances only, prove the figure is a rectangle, and find its perimeter, area and the length of each diagonal.

SOLUTION

Step 1: Given: four points in cyclic order. Needed: prove rectangle; perimeter, area, diagonals.

Step 2: KL: y fixed at 1, length |5 - 1| = 4 (horizontal).

Step 3: LM: x fixed at 5, length |4 - 1| = 3 (vertical).

Step 4: MN: y fixed at 4, length |5 - 1| = 4 (horizontal). NK: x fixed at 1, length |4 - 1| = 3 (vertical).

Step 5: Opposite sides are equal, and every side is axis-parallel, so adjacent sides meet at right angles: the figure is a rectangle.

Step 6: Diagonals: KM = $\sqrt{((5 - 1)^2 + (4 - 1)^2)} = \sqrt{(16 + 9)} = \sqrt{25} = 5$. LN = $\sqrt{((1 - 5)^2 + (4 - 1)^2)} = \sqrt{(16 + 9)} = 5$. Equal diagonals confirm the rectangle property.

Step 7: Perimeter = 2(4 + 3) = 14 units. Area = 4 x 3 = 12 square units. Both diagonals are 5.

Hinglish: *Ye question rectangle-proof ka full kit hai: (1) opposite sides barabar, (2) sab sides axis-parallel toh right angles automatic, (3) diagonals barabar - teeno checks ho gaye. 3-4-5 triple yahan diagonals mein chhupa hai! Perimeter 14, area 12, diagonal 5. Aise structured answers examiner ko bahut pasand aate hain kyunki har property alag line par verified hai.*

Answer: It is a rectangle (equal opposite sides, axis-parallel sides, equal diagonals of length 5); perimeter 14 units, area 12 sq units.

Q40 · High · Concept: Quadrant-restricted distances (T1/T4)

Q40. A point P lies in Quadrant II. Its perpendicular distance from the x-axis is 5 and from the y-axis is 3. Write the coordinates of P and find its distance from the origin. If a twin point P' in Quadrant I has the same axis distances, what is the length PP'?

SOLUTION

Step 1: Given: Quadrant II point with axis distances 5 (from x-axis) and 3 (from y-axis). Needed: P, OP, and PP'.

Step 2: Distance from the x-axis is the size of the y-coordinate: |y| = 5. Distance from the y-axis is |x| = 3.

Step 3: Quadrant II means x negative, y positive. So x = -3 and y = +5, giving P(-3, 5).

Step 4: OP = $\sqrt{((-3)^2 + 5^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 5: P' in Quadrant I with the same distances: x = +3, y = +5, so P'(3, 5).

Step 6: PP' joins (-3, 5) to (3, 5): same y, so PP' = |3 - (-3)| = 6 (horizontal segment).

Step 7: Note the traps avoided: the distance from the x-axis (5) became the y-coordinate, and the Quadrant II condition fixed both signs.

» Why this step? Translating 'distance from the x-axis' into 'absolute y-value' is the key step; students who mix this up put 3 and 5 in the wrong slots.

Hinglish: *Sabse pehle translation: x-axis se doori 5 matlab |y| = 5; y-axis se doori 3 matlab |x| = 3. Ab quadrant II ka rule lagao - x negative, y positive: P(-3, 5). OP = √34 (koi triple nahi bana, root form chal raha hai). Quadrant I wala twin (3, 5). Dono ka y same, toh distance sirf x ka gap = 6. Ye question pure sign-discipline ka test hai - ek sign ulta kiya toh poora answer flip.*

Answer: P(-3, 5); OP = √34; P'(3, 5); PP' = 6 units.

Q41 · HOTS · Concept: Two intersection points on an axis (T5)

Q41. Find ALL points on the x-axis that lie at distance 10 from the point C(3, -6). Explain geometrically why there must be two answers.

SOLUTION

Step 1: Given: points (x, 0) at distance 10 from C(3, -6). Needed: all such x.

Step 2: Distance equation: $\sqrt{((x - 3)^2 + (0 - (-6))^2)} = 10$.

Step 3: Square both sides: $(x - 3)^2 + 36 = 100$.

Step 4: So $(x - 3)^2 = 64$, giving $x - 3 = 8$ or $x - 3 = -8$.

Step 5: Thus $x = 11$ or $x = -5$. The points are (11, 0) and (-5, 0).

Step 6: Verify: from (11, 0) to (3, -6): $\sqrt{(64 + 36)} = \sqrt{100} = 10$. From (-5, 0) to (3, -6): $\sqrt{(64 + 36)} = 10$. Both valid.

Step 7: Geometry: a circle of radius 10 centred at C(3, -6) cuts the x-axis at exactly two points because the centre sits 6 units above the axis, which is less than the radius 10 - the circle dips below the axis on both sides of $x = 3$.

» **Why this step?** The equation $(x - 3)^2 = 64$ is a disguised two-case problem; the geometric picture (circle cutting a line) tells you in advance how many answers to expect.

Hinglish: *Circle-axis ka milaap socho: C se radius 10 ka circle banao, wo x-axis ko kata kyun? Kyunki C axis se sirf 6 door hai aur radius 10 hai - circle axis ke paar niche tak jaata hai, dono taraf se cut karta hai. Equation mein bhi wahi dikh raha hai: $(x-3)^2 = 64$ ke do roots: $x = 11$ aur $x = -5$. Jab bhi square = positive number dikhe, do answers ready rakho. Aur agar centre axis se doori radius se ZYADA hoti, toh zero answers aate - wo case bhi soch ke dekho!*

Answer: (11, 0) and (-5, 0); two answers because the circle of radius 10 around C(3, -6) crosses the x-axis twice.

Q42 · HOTS · Concept: Circumcentre discovery (T10/T14)

Q42. For the right triangle with vertices A(0, 0), B(8, 0) and C(8, 6), find the point that is equidistant from all three vertices. Where does it sit, and what pattern do you notice?

SOLUTION

Step 1: Given: right triangle with legs on the axes... (adapted: right angle at B since AB is horizontal and BC is vertical). Needed: the common-equidistance point.

Step 2: AB = 8 (horizontal), BC = 6 (vertical), so the right angle is at B(8, 0), and AC is the hypotenuse.

Step 3: $AC = \sqrt{((8 - 0)^2 + (6 - 0)^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 4: Claim: the midpoint of the hypotenuse is equidistant from all three vertices. Midpoint M of AC = ((0 + 8)/2, (0 + 6)/2) = (4, 3).

Step 5: Check $MA = \sqrt{(4^2 + 3^2)} = 5$. $MC = \sqrt{((8 - 4)^2 + (6 - 3)^2)} = \sqrt{(16 + 9)} = 5$.

Step 6: $MB = \sqrt{((8 - 4)^2 + (0 - 3)^2)} = \sqrt{(16 + 9)} = 5$. All three distances are 5 - verified.

Step 7: Pattern: in a right triangle, the midpoint of the hypotenuse is equidistant from all three vertices (it is the centre of the circle passing through the three corners). This is the coordinate-proof version of a famous theorem.

Hinglish: *Ye question ek theorem khud discover karwa raha hai! Right angle B par hai (horizontal + vertical legs), hypotenuse AC = 10. Uska midpoint (4, 3) lo - teeno vertices se distance 5, 5, 5. Matlab: right triangle mein hypotenuse ka midpoint hi wo magical point hai jo teeno corners se barabar door baithta hai (circumcentre). 3-4-5*

triple yahan bhi kaam kar gaya. Ye result yaad rakho - contest maths mein baar-baar kaam aayega.

Answer: M(4, 3), the midpoint of the hypotenuse, is equidistant (distance 5) from all three vertices.

Q43 · HOTS · Concept: Sum-of-distances locus (T14)

Q43. A(-4, 0) and B(4, 0) are two fixed points. P(0, y) lies on the y-axis such that PA + PB = 10. Find all possible y. Why is y = 0 NOT a solution?

SOLUTION

Step 1: Given: PA + PB = 10 with P on the y-axis. Needed: y values.

Step 2: PA = $\sqrt{((0 + 4)^2 + y^2)} = \sqrt{(16 + y^2)}$. By symmetry PB has the same value $\sqrt{(16 + y^2)}$.

Step 3: So the condition reads $2\sqrt{(16 + y^2)} = 10$, hence $\sqrt{(16 + y^2)} = 5$.

Step 4: Square: $16 + y^2 = 25$, so $y^2 = 9$, giving y = 3 or y = -3.

Step 5: Verify y = 3: PA = $\sqrt{(16 + 9)} = 5$, PB = 5, total 10. Same for y = -3. Both work.

Step 6: Why not y = 0: then PA = PB = 4 and the sum is 8, which is short of 10 - P at the origin sits strictly between A and B, so PA + PB just equals AB = 8, the minimum possible value. To reach 10 the point must climb off the line.

Hinglish: *Symmetry dekho: A aur B origin ke dono taraf barabar door hain, isliye PA = PB hoga hi. Toh 2 x distance = 10, distance = 5. 16 + y^2 = 25 se y = plus-minus 3. Aur y = 0 kyun nahi: origin par P ke paas PA + PB = AB hi hota hai = 8 - shortest possible. 10 chahiye toh P ko upar ya neeche jhukna padega (y = 3 ya -3). Ye question aage ke classes mein ellipse ka darwaza hai - PA + PB constant wala idea.*

Answer: y = 3 or y = -3; y = 0 fails because PA + PB = AB = 8 there, the minimum, not 10.

Q44 · HOTS · Concept: Box method for area (T12/T14)

Q44. A triangle has vertices A(0, 0), B(1, 2) and C(3, 1). Show that it is isosceles, and find its area using the box (enclosing rectangle) method.

SOLUTION

Step 1: Given: three lattice points. Needed: isosceles proof and area.

Step 2: AB = $\sqrt{(1^2 + 2^2)} = \sqrt{5}$. BC = $\sqrt{((3 - 1)^2 + (1 - 2)^2)} = \sqrt{(4 + 1)} = \sqrt{5}$. AC = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$.

Step 3: AB = BC = $\sqrt{5}$, so the triangle is isosceles (apex at B).

Step 4: Box method: enclose the triangle in the smallest axis-aligned rectangle: corners (0, 0), (3, 0), (3, 2), (0, 2). Box area = 3 x 2 = 6.

Step 5: The box minus the triangle leaves three right triangles: T1 below AB with legs 1 and 2 (area 1), T2 right of BC with legs 2 and 1 (area 1), T3 above AC with legs 3 and 1 (area 1.5).

Step 6: Area of ABC = 6 - 1 - 1 - 1.5 = 2.5 square units.

Step 7: Cross-check by another triangle with same side data is not needed; instead sanity-check magnitude: the triangle fits inside a 3 x 2 box, so 2.5 is plausible.

» Why this step? When the triangle is not right-angled and the base-height pair is not obvious, the box method computes area using only right triangles - fully within Chapter 1 tools.

Hinglish: *Jab triangle tedha ho aur base-height na dikhe, box method use karo: triangle ko chhote rectangle mein band karo, rectangle ka area lo, aur charo taraf ke right triangles ghata do. 6 - 1 - 1 - 1.5 = 2.5. Ye technique squares ke root wale triangles ka area nikalne ka desi jugaad hai - bina kisi naye formula ke. Aur dekho, AB = BC = √5 se isosceles bhi proof ho gaya. Do kaam ek question mein.*

Answer: Isosceles (AB = BC = √5); area = 2.5 square units via the box method.

Q45 · HOTS · Concept: Chord comparison on a circle (T10/T4)

Q45. Three points A(3, 4), B(-4, 3) and C(0, -5) all lie on the circle of radius 5 centred at the origin (verify this). Find the lengths of the chords AB, BC and CA, and state which chord is the longest.

SOLUTION

Step 1: Given: three points, circle radius 5. Needed: verify membership, three chord lengths, longest chord.

Step 2: Membership: A: 9 + 16 = 25. B: 16 + 9 = 25. C: 0 + 25 = 25. All equal to $r^2 = 25$, so all three lie on the circle.

Step 3: AB = $\sqrt{((-4 - 3)^2 + (3 - 4)^2)} = \sqrt{(49 + 1)} = \sqrt{50} = 5\sqrt{2}$ (approximately 7.07).

Step 4: BC = $\sqrt{((0 + 4)^2 + (-5 - 3)^2)} = \sqrt{(16 + 64)} = \sqrt{80} = 4\sqrt{5}$ (approximately 8.94).

Step 5: CA = $\sqrt{((3 - 0)^2 + (4 + 5)^2)} = \sqrt{(9 + 81)} = \sqrt{90} = 3\sqrt{10}$ (approximately 9.49).

Step 6: Comparing exact squares: 90 > 80 > 50, so CA > BC > AB.

Step 7: The longest chord is CA with length $3\sqrt{10}$. (No chord can exceed the diameter 10, and indeed all three are below 10.)

Hinglish: *Pehle membership - teeno 25 de rahe hain, sab circle par. Phir chords: √50, √80, √90. Compare karne ka clean tareeka: squares compare karo (50 < 80 < 90), roots ki zaroorat nahi. Longest CA = 3√10 approx 9.49. Aur ek boundary-check: diameter 10 hai, koi chord 10 se bada nahi - hamara answer us limit ke andar hi hai. Yahi internal consistency marks dilati hai.*

Answer: All three lie on the circle; AB = $5\sqrt{2}$, BC = $4\sqrt{5}$, CA = $3\sqrt{10}$; the longest chord is CA.

Q46 · HOTS · Concept: Parallelogram via diagonal midpoints (T8/T6)

Q46. A(1, 2), B(4, 3) and C(6, 7) are three vertices of parallelogram ABCD taken in order. Find the fourth vertex D using the fact that the diagonals of a parallelogram bisect each other, and verify your answer.

SOLUTION

Step 1: Given: three consecutive vertices of a parallelogram. Needed: D.

Step 2: Diagonals of a parallelogram bisect each other, so midpoint of AC must equal midpoint of BD.

Step 3: Midpoint of AC = ((1 + 6)/2, (2 + 7)/2) = (3.5, 4.5).

Step 4: Let D = (x, y). Midpoint of BD = ((4 + x)/2, (3 + y)/2). Set equal to (3.5, 4.5).

Step 5: From x: (4 + x)/2 = 3.5 gives x = 3. From y: (3 + y)/2 = 4.5 gives y = 6.

Step 6: So D = (3, 6).

Step 7: Verify as a parallelogram by sides: AB = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$, DC = $\sqrt{((6 - 3)^2 + (7 - 6)^2)} = \sqrt{(9 + 1)} = \sqrt{10}$; BC = $\sqrt{(2^2 + 4^2)} = \sqrt{20} = 2\sqrt{5}$, AD = $\sqrt{((3 - 1)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20}$. Opposite sides equal - consistent.

» Why this step? The diagonal-midpoint method converts a geometry condition into two tiny linear equations - far safer than guessing a translation vector.

Hinglish: *Parallelogram ka chautha vertex nikalne ka RULE: diagonals ka midpoint same hota hai. AC ka midpoint lo, phir BD ka midpoint uske barabar rakho - do chhote equations mil gaye: x = 3, y = 6. Verification mein opposite sides bhi barabar nikli (√10 aur 2√5 ke pairs). Ye midpoint-trick Q31 (trisection) ke saath milke midpoint-based toolkit complete karta hai: aage class 10 mein yehi idea hi midpoint formula ka base banega.*

Answer: D = (3, 6); verified since AB = DC = $\sqrt{10}$ and BC = AD = $2\sqrt{5}$.

Q47 · HOTS · Concept: Grid-walk vs direct distance (T11/T4)

Q47. A delivery robot moves on a grid of unit squares. It must travel from station S(1, 1) to station T(7, 4). (i) What is the length of the shortest path using only horizontal and vertical moves? (ii) What is the straight-line distance? (iii) A second robot flies directly. How much distance does it save compared to the ground robot?

SOLUTION

Step 1: Given: grid unit 1, S(1, 1) to T(7, 4). Needed: street length, direct length, saving.

Step 2: Horizontal gap = 7 - 1 = 6 units; vertical gap = 4 - 1 = 3 units.

Step 3: Ground robot (only axis moves): 6 + 3 = 9 units. Every such path has exactly this length - no order of moves can beat it.

Step 4: Flying robot (straight line): $\sqrt{(6^2 + 3^2)} = \sqrt{(36 + 9)} = \sqrt{45} = 3\sqrt{5}$ (approximately 6.71 units).

Step 5: Saving = $9 - 3\sqrt{5}$ (exactly), approximately 9 - 6.71 = 2.29 units.

Step 6: Keep the exact answer $9 - 3\sqrt{5}$ in the final statement; the decimal is only for intuition.

Hinglish: *Grid-walk ka funda: horizontal steps + vertical steps hamesha fix hote hain (yahan 6 + 3 = 9), chahe kisi bhi order mein chalo. Direct distance Pythagoras se $3\sqrt{5}$. Saving $9 - 3\sqrt{5}$ approx 2.3 units. Ye question Q26 (city auto) ka bhai hai - wahan 200 m blocks the, yahan 1 unit. Same skeleton, naye numbers - yahi 'question structure' samajhna original practice hai, ratna nahi.*

Answer: (i) 9 units (ii) $3\sqrt{5}$ units (approx 6.71) (iii) saving $9 - 3\sqrt{5}$ approx 2.29 units.

Q48 · HOTS · Concept: Equally spaced collinear chain (T9/T8)

Q48. P(1, 2), Q(4, 6) and R(7, 10) are three points. Show they are collinear and equally spaced. Then find the point T on the same line, beyond R, with QT = 10.

SOLUTION

Step 1: Given: three points. Needed: collinearity, equal spacing, then T.

Step 2: PQ = $\sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(9 + 16)} = 5$.

Step 3: QR = $\sqrt{((7 - 4)^2 + (10 - 6)^2)} = \sqrt{(9 + 16)} = 5$.

Step 4: PR = $\sqrt{((7 - 1)^2 + (10 - 2)^2)} = \sqrt{(36 + 64)} = 10$.

Step 5: Since PR = PQ + QR = 10, the points are collinear, and PQ = QR = 5 shows Q is exactly the midpoint - equally spaced.

Step 6: The repeated step from P to Q is (+3, +4). The same step continues from Q to R, and one more step lands at T = (7 + 3, 10 + 4) = (10, 14).

Step 7: Check QT: from Q(4, 6) to T(10, 14) is two steps: 2 x 5 = 10. Verified.

» Why this step? A constant coordinate-step (+3, +4) between consecutive points is the signature of equal spacing on a line - recognise the pattern and distances become trivial.

Hinglish: *Pattern-spotting question! Har agle point mein (+3, +4) jud raha hai - 3-4-5 triple ka repeated use. PQ = QR = 5, PR = 10: collinear + equally spaced confirmed. Ab 'beyond R with QT = 10' ka matlab: Q se 2 step aage = R se 1 step aage = T(10, 14). Verify kiya QT = 10. Aise 'chain' questions mein step-pakadna hi asli skill hai, har baar distance formula chalana nahi.*

Answer: Collinear with PQ = QR = 5, PR = 10; T = (10, 14) with QT = 10 verified.

Q49 · HOTS · Concept: Chord and centre distance (T10/T8)

Q49. A circle of radius 10 is centred at the origin. A and B(−8, 6) are two points on it, with A(6, 8). Find the length of chord AB and the perpendicular distance of the chord from the centre. Verify the right-triangle relation between them.

SOLUTION

Step 1: Given: circle $x^2 + y^2 = 100$, chord joining A(6, 8) and B(-8, 6). Needed: chord length, distance from centre, consistency check.

Step 2: First confirm membership: A: 36 + 64 = 100. B: 64 + 36 = 100. Both on the circle.

Step 3: Chord AB = $\sqrt{((-8 - 6)^2 + (6 - 8)^2)} = \sqrt{(196 + 4)} = \sqrt{200} = 10\sqrt{2}$.

Step 4: Midpoint M of the chord = ((6 + (-8))/2, (8 + 6)/2) = (-1, 7).

Step 5: The line from the centre to the midpoint of a chord is perpendicular to the chord, so the required distance is OM = $\sqrt{((-1)^2 + 7^2)} = \sqrt{50} = 5\sqrt{2}$.

Step 6: Right-triangle check: half-chord = $5\sqrt{2}$, OM = $5\sqrt{2}$, radius = 10. Indeed $(5\sqrt{2})^2 + (5\sqrt{2})^2 = 50 + 50 = 100 = 10^2$.

Step 7: Everything is consistent - the chord, its distance from the centre and the radius obey the Baudhayana-Pythagoras relation.

Hinglish: *Do kaam ek saath: chord length (distance formula se $10\sqrt{2}$) aur centre se doori. Fundaa: chord ke midpoint se centre ko jodo - wo line chord par perpendicular hoti hai, toh wahi distance hai. OM = $\sqrt{50} = 5\sqrt{2}$. Phir master-check: (half-chord)² + (distance)² = radius². 50 + 50 = 100. Perfect. Ye relation har circle-chord question ka backbone hai - yahin se 'perpendicular from centre bisects chord' wali theory aati hai.*

Answer: AB = $10\sqrt{2}$; distance from centre = $5\sqrt{2}$; verified since $(5\sqrt{2})^2 + (5\sqrt{2})^2 = 100 = \text{radius}^2$.

Q50 · HOTS · Concept: Full square proof chain (T14)

Q50. Prove that the points P(1, 7), Q(4, 2), R(-1, -1) and S(-4, 4), taken in this order, are the vertices of a square. You must verify ALL necessary properties using the distance formula and midpoints.

SOLUTION

Step 1: Given: four points in cyclic order. Needed: full square proof.

Step 2: PQ = $\sqrt{((4 - 1)^2 + (2 - 7)^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 3: QR = $\sqrt{((-1 - 4)^2 + (-1 - 2)^2)} = \sqrt{(25 + 9)} = \sqrt{34}$.

Step 4: RS = $\sqrt{((-4 + 1)^2 + (4 + 1)^2)} = \sqrt{(9 + 25)} = \sqrt{34}$.

Step 5: SP = $\sqrt{((1 + 4)^2 + (7 - 4)^2)} = \sqrt{(25 + 9)} = \sqrt{34}$. All four sides equal: it is at least a rhombus-shaped candidate.

Step 6: Diagonals: PR = $\sqrt{((-1 - 1)^2 + (-1 - 7)^2)} = \sqrt{(4 + 64)} = \sqrt{68} = 2\sqrt{17}$. QS = $\sqrt{((-4 - 4)^2 + (4 - 2)^2)} = \sqrt{(64 + 4)} = \sqrt{68}$. Diagonals are equal.

Step 7: Bisecting check: midpoint of PR = ((1 + (-1))/2, (7 + (-1))/2) = (0, 3). Midpoint of QS = ((4 + (-4))/2, (2 + 4)/2) = (0, 3). Diagonals bisect each other, so the figure is a parallelogram.

Step 8: A parallelogram with all sides equal is a rhombus; a rhombus with equal diagonals is a square. Hence PQRS is a square.

Step 9: Optional right-angle cross-check: $PR^2 = 68$ and $PQ^2 + QR^2 = 34 + 34 = 68$, so the angle at Q (opposite diagonal PR) is 90 degrees by the converse of Pythagoras.

» Why this step? The clean chain for proving a square with distances only: sides equal (rhombus candidate) + diagonals bisect (parallelogram) + diagonals equal (square). Each link uses only Chapter 1 tools.

Hinglish: *Ye Chapter 1 ka final boss hai. Strategy 3-layer ki hai: (1) chaaron sides √34 - rhombus material, (2) diagonals ke midpoints dono (0, 3) - parallelogram confirm, (3) diagonals dono √68 - square confirm. Bonus check: 68 = 34 + 34 se right angle bhi mil gaya. Notice karo kitna systematic hai - 'sides equal + diagonals bisect + diagonals equal' ye teen-step recipe kisi bhi square-proof question mein chalegi. Ye last question mein jaan-bujh ke negative coordinates aur non-axis orientation rakha hai taaki tum axes par dependent na raho.*

Answer: PQRS is a square: all sides √34, diagonals equal (2√17) and bisecting each other at (0, 3), with right angles confirmed by 68 = 34 + 34.

1.4 Chapter MCQs (60 questions, answers inline)

Each MCQ lists the options, the correct answer, a full explanation, the trap analysis for the tempting wrong options, and a Hinglish note. Options are engineered from real student mistakes, so think before you pick.

MCQ 1 · Normal · Concept: Quadrant signs

1. The point (-7, 4) lies in which quadrant?

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Correct Answer: (B)

x = -7 is negative and y = +4 is positive. The sign pattern (-, +) is exactly the definition of Quadrant II (above the x-axis, left of the y-axis).

Why (A) is wrong: (+, +) would need BOTH positive; here x is negative.

Why (C) is wrong: Quadrant III is (-, -) - y must also be negative.

Why (D) is wrong: Quadrant IV is (+, -) - both signs are wrong here.

Hinglish: *Pehla number minus (left), doosra plus (upar) - ye Quadrant II ka signature hai. Bas signs match karo, answer mil gaya.*

MCQ 2 · Normal · Concept: Points on axes

2. The point on the y-axis lying 6 units above the origin has coordinates:

A. (6, 0)
B. (0, 6)
C. (-6, 0)
D. (0, -6)

Correct Answer: (B)

On the y-axis the x-coordinate must be 0. Above the origin means positive y. Hence (0, 6).

Why (A) is wrong: (6, 0) is on the x-axis, not the y-axis.

Why (D) is wrong: (0, -6) is 6 units BELOW the origin.

Hinglish: *Y-axis = pehla coordinate zero. Upar = positive. (0, 6). Itni si baat, par (6, 0) likhne wale har saal milte hain!*

MCQ 3 · Normal · Concept: Coordinate meaning

3. The perpendicular distance of the point (-4, 9) from the y-axis is:

A. 9 units
B. 4 units
C. -4 units
D. √97 units

Correct Answer: (B)

Distance from the y-axis is given by the size of the x-coordinate: |x| = |-4| = 4. Distances are never negative.

Why (A) is wrong: 9 is the distance from the x-axis, not the y-axis.

Why (C) is wrong: Distances are always non-negative; -4 is a coordinate, not a distance.

Why (D) is wrong: √97 is the distance from the origin.

Hinglish: *y-axis se doori = |x|. Minus sirf side batata hai (left), distance toh 4 hi hai. Distance kabhi negative nahi hota - option C khatam. √97 toh origin se distance hai ($4^2 + 9^2$).*

MCQ 4 · Normal · Concept: Vertical distance

4. The distance between the points (3, -2) and (3, 7) is:

A. 5 units
B. 9 units
C. √85 units
D. 10 units

Correct Answer: (B)

Same x-coordinate means a vertical segment. Distance = |7 - (-2)| = |7 + 2| = 9.

Why (A) is wrong: 5 confuses the gap with the smaller coordinate.

Why (C) is wrong: √85 results from a wrong application of the distance formula.

Hinglish: *x same hai toh seedha y ka gap: 7 - (-2) = 9. Minus-minus plus bana - yahi sabse common slip hai. √85 aata hai agar log dono differences ghalat jod dein.*

MCQ 5 · Normal · Concept: Distance formula 6-8-10

5. The distance between (1, -2) and (7, 6) is:

A. 10 units
B. 12 units
C. √84 units
D. 14 units

Correct Answer: (A)

Differences: 7 - 1 = 6 and 6 - (-2) = 8. Distance = √(36 + 64) = √100 = 10, the 6-8-10 triple.

Why (B) is wrong: 12 would come from adding instead of applying Pythagoras.

Why (C) is wrong: √84 comes from miscomputing the y-difference as 4 or 8-(-2) as something else.

Hinglish: *6 aur 8 dikh rahe hain? 3-4-5 ka do guna - seedha 10. Triple yaad rakho toh square karne ki bhi zaroorat nahi.*

MCQ 6 · Normal · Concept: Conceptual: free coordinate

6. For how many values of k is the distance between (k, 3) and (k, 8) equal to 5?

A. Exactly one value of k
B. Exactly two values of k
C. All real values of k
D. No value of k

Correct Answer: (C)

Both points have the same x-coordinate k, so the distance is |8 - 3| = 5 regardless of k. Every real k works - there are infinitely many such pairs.

Why (A) is wrong: There is no equation restricting k.

Why (B) is wrong: The distance does not depend on k at all.

Hinglish: *Ye trick question hai! Distance dono points ke y-gap se banti hai jo hamesha 5 hai - k ka kuch lena dena hi nahi. Isliye saare real k chalega. Log sochte hain k = 5 hoga - nahi bhai, k toh bas dono ko same rakhta hai.*

MCQ 7 · Normal · Concept: Distance from origin

7. The distance of the point (-6, 8) from the origin is:

A. 2 units
B. 10 units
C. 14 units
D. 100 units

Correct Answer: (B)

$\sqrt{((-6)^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$. Signs die inside squares; the 6-8-10 triple does the rest.

Why (A) is wrong: 2 = |-6 - 8| is a meaningless subtraction of coordinates.

Why (C) is wrong: Adding coordinate sizes is not a distance.

Why (D) is wrong: 100 is the squared distance, not the distance.

Hinglish: *Minus square mein mar jaata hai: 36 + 64 = 100, root 10. 14 wo log chunte hain jo -6 + 8 ko distance samajh lete hain - aisa formula exist hi nahi karta.*

MCQ 8 · Normal · Concept: Half-turn symmetry

8. If P(a, b) lies in Quadrant II, then the point (-a, -b) lies in:

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Correct Answer: (D)

In Q II, a is negative and b is positive. So -a is positive and -b is negative: signs (+, -), which is Quadrant IV. Flipping both signs reflects the point through the origin (a half-turn).

Why (A) is wrong: Both signs must flip, not stay one-negative-one-positive.

Why (B) is wrong: Same quadrant is impossible after a half-turn.

Hinglish: *Q II mein a negative, b positive. Dono ke sign palte toh (+, -) = Q IV. Yaad rakho: (x, y) aur (-x, -y) hamesha opposite quadrants mein hote hain - origin ke through mirror.*

MCQ 9 · Normal · Concept: Reflection in y-axis

9. The reflection of the point (-3, -5) in the y-axis is:

A. (3, -5)
B. (3, 5)
C. (-3, 5)
D. (-3, -5)

Correct Answer: (A)

Reflection in the y-axis flips only the sign of x: (-3, -5) becomes (3, -5). The y-coordinate is untouched.

Why (B) is wrong: (3, 5) flips both signs - that is reflection in BOTH axes.

Why (C) is wrong: (-3, 5) flips y only - that is the x-axis reflection.

Hinglish: *y-axis mein mirror = sirf x ka sign ulta. (-3, -5) ka x minus tha, ab plus: (3, -5). y ko mat chhedo - wahi sabse common galti hai.*

MCQ 10 · Normal · Concept: Reflection in x-axis

10. The reflection of (2, 7) in the x-axis, followed by reflection of the image in the y-axis, gives:

A. (-2, -7)
B. (2, -7)
C. (-2, 7)
D. (7, -2)

Correct Answer: (A)

x-axis reflection: (2, 7) to (2, -7). Then y-axis reflection flips x: (2, -7) to (-2, -7). Two axis reflections = half-turn = both signs flip.

Why (B) is wrong: Only one reflection was performed.

Why (C) is wrong: Wrong axis was flipped.

Why (D) is wrong: Coordinates were swapped - reflections never swap coordinates.

Hinglish: *Do reflections ki chain = dono signs flip. (2, 7) se seedha (-2, -7). Shortcut yaad rakho: x-axis phir y-axis reflect karna = origin ke around 180 degree ghumaana.*

MCQ 11 · Normal · Concept: Midpoint

11. The midpoint of the segment joining (-3, 8) and (5, -2) is:

A. (1, 3)
B. (2, 6)
C. (1, 5)
D. (4, 3)

Correct Answer: (A)

Midpoint = ((-3 + 5)/2, (8 + (-2))/2) = (2/2, 6/2) = (1, 3).

Why (B) is wrong: (2, 6) is the sum of the coordinates without halving.

Why (C) is wrong: The y-average was computed as (8 - 2)/2 = 5 by dropping a sign.

Why (D) is wrong: Adds only the x-pair wrongly.

Hinglish: *Average lo dono coordinates ka: (-3 + 5)/2 = 1 aur (8 - 2)/2 = 3. (2, 6) wo galti hai jab divide karna bhool jaate ho.*

MCQ 12 · Normal · Concept: Reverse midpoint

12. If M(2, -1) is the midpoint of A(-4, 3) and B(x, y), then B is:

A. (8, -5)
B. (-10, 7)
C. (8, 5)
D. (0, 1)

Correct Answer: (A)

Undo the average: x = 2(2) - (-4) = 8 and y = 2(-1) - 3 = -5. So B = (8, -5).

Why (B) is wrong: Adds instead of subtracting the known endpoint.

Why (C) is wrong: y computed as -2 + 3.

Why (D) is wrong: Halves something unnecessarily.

Hinglish: *Midpoint se endpoint nikalna = 2 x midpoint minus known point. x: 4 + 4 = 8. y: -2 - 3 = -5. Sign ka khayal: 3 ko MINUS karna hai, plus nahi.*

MCQ 13 · Normal · Concept: Collinearity

13. Which set of three points is collinear?

A. (0, 0), (2, 1), (4, 2)
B. (0, 1), (2, 2), (5, 4)
C. (1, 1), (2, 3), (4, 6)
D. (0, 0), (1, 2), (3, 7)

Correct Answer: (A)

In option A each step from one point to the next is (+2, +1) - a constant step means equal slope, so the points lie on one line (y = x/2). The other sets change the step pattern, so they bend.

Why (B) is wrong: Steps (+2, +1) then (+3, +2) - slopes 1/2 vs 2/3 differ.

Why (C) is wrong: Slopes 2 vs 3/2 differ.

Why (D) is wrong: Slopes 2 vs 5/2 differ.

Hinglish: *Collinear ka fastest check: consecutive points ke beech ka STEP same hai ya nahi. Option A mein har baar (+2, +1) - perfect line. Baaki options mein step badal jaata hai, matlab line mudti hai.*

MCQ 14 · Normal · Concept: Circle membership

14. Which point lies ON the circle $x^2 + y^2 = 10$ with centre the origin?

A. (1, 3)
B. (2, 2)
C. (1, 2)
D. (3, 3)

Correct Answer: (A)

Check $x^2 + y^2$: (1, 3) gives 1 + 9 = 10 exactly. (2, 2) gives 8 (inside), (1, 2) gives 5 (inside), (3, 3) gives 18 (outside).

Why (B) is wrong: 8 < 10 means inside, not on.

Why (C) is wrong: 5 is far from 10.

Why (D) is wrong: 18 > 10 means outside.

Hinglish: *Equation mein plug karo, 10 aana chahiye. (1, 3): 1 + 9 = 10 perfect. Baaki sab 10 nahi de rahe - andar ya bahar.*

MCQ 15 · Normal · Concept: 5-12-13 triple

15. The distance of (5, -12) from the origin is:

A. 7 units
B. 13 units
C. 17 units
D. 169 units

Correct Answer: (B)

$\sqrt{(25 + 144)} = \sqrt{169} = 13$. The 5-12-13 triple, again with a harmless minus sign.

Why (A) is wrong: 5 + (-12) size subtraction - meaningless.

Why (C) is wrong: 17 = 5 + 12 (adding instead of Pythagoras).

Why (D) is wrong: 169 is 13 squared.

Hinglish: *5-12-13 triple ratt lo. Minus ko ignore karo - square ke baad koi matlab nahi. 169 square hai, root nahi - aisi chhoti galti bahut log karte hain.*

MCQ 16 · Normal · Concept: Symbolic distance

16. The distance between the points (a, 0) and (0, a) (a > 0) is:

A. 2a units
B. $a\sqrt{2}$ units
C. a units
D. 0 units

Correct Answer: (B)

Differences: a - 0 = a and 0 - a = -a. Distance = $\sqrt{(a^2 + a^2)} = \sqrt{(2a^2)} = a\sqrt{2}$.

Why (A) is wrong: 2a would be a + a, but legs add only through squares.

Why (C) is wrong: a is just one leg.

Why (D) is wrong: The points are distinct unless a = 0.

Hinglish: *Dono differences ka size a hai - isosceles right triangle ka hypotenuse = $a\sqrt{2}$. 2a karna Pythagoras ko addition samajhna hai - bada classic mistake.*

MCQ 17 · Normal · Concept: Perimeter of right triangle

17. The perimeter of the triangle with vertices (0, 0), (4, 0), (0, 3) is:

A. 7 units
B. 12 units
C. 10 units
D. 6 units

Correct Answer: (B)

Legs are 4 and 3; hypotenuse = 5 (3-4-5). Perimeter = 4 + 3 + 5 = 12.

Why (A) is wrong: 3 + 4 only - forgot the hypotenuse.

Why (C) is wrong: Confused with a different triple.

Why (D) is wrong: Half the perimeter.

Hinglish: *Axis par baithe legs 4 aur 3, hypotenuse 5. Sab JODO: 12. 7 sirf do legs ka sum hai - hypotenuse bhool gaye.*

MCQ 18 · Normal · Concept: Area of right triangle

18. The area of the triangle with vertices (0, 0), (8, 0) and (0, 6) is:

A. 48 sq units
B. 24 sq units
C. 14 sq units
D. 10 sq units

Correct Answer: (B)

Right angle at the origin with legs 8 and 6. Area = (1/2)(8)(6) = 24.

Why (A) is wrong: Missing the factor 1/2.

Why (C) is wrong: 8 + 6 - a sum, not an area.

Why (D) is wrong: Applied some ratio incorrectly.

Hinglish: *Right triangle ka area = aadha x base x height = 24. 48 wo galti hai jab 1/2 multiply karna bhool jaate ho.*

MCQ 19 · Normal · Concept: Assertion-reason

19. Assertion (A): The distance of a point (x, y) from the origin is $\sqrt{(x^2 + y^2)}$. Reason (R): The Baudhayana-Pythagoras Theorem gives hypotenuse = √(sum of squares of legs). Choose the correct option:

A. Both A and R are true and R explains A
B. Both A and R are true but R does not explain A
C. A is true, R is false
D. A is false, R is true

Correct Answer: (A)

The distance formula from the origin is exactly Pythagoras applied to the legs x and y. The assertion is true and the reason is the actual justification behind it.

Why (B) is wrong: R genuinely explains A here - do not dismiss the connection.

Why (C) is wrong: R is a correct statement of the theorem.

Why (D) is wrong: A is the standard distance-from-origin result.

Hinglish: *A sahi hai, aur wo sahi ISLIYE hai kyunki origin se jaane wala segment right triangle ka hypotenuse hai jiske legs x aur y hain. R wahi theorem hai - toh R, A ko explain karta hai. Option A.*

MCQ 20 · Normal · Concept: Horizontal segment

20. The line joining (-5, 2) and (5, 2) is parallel to:

A. the x-axis
B. the y-axis
C. both axes
D. neither axis

Correct Answer: (A)

Both points have y = 2, so the segment is horizontal, i.e. parallel to the x-axis (and perpendicular to the y-axis).

Why (B) is wrong: Same y means horizontal, not vertical.

Why (C) is wrong: A non-degenerate segment cannot be parallel to both perpendicular axes.

Hinglish: *y same = horizontal line = x-axis ke parallel. Same x hota toh y-axis ke parallel hota.*

MCQ 21 · Normal · Concept: Equidistant on x-axis

21. The point (x, 0) is equidistant from (2, 3) and (6, 3). Then x equals:

A. 4
B. 2
C. 8
D. 3

Correct Answer: (A)

$(x - 2)^2 + 9 = (x - 6)^2 + 9$. The 9s cancel: $(x - 2)^2 = (x - 6)^2$, so x - 2 = plus/minus (x - 6). The negative-sign branch gives a contradiction (0 = -8... no solution from that branch), the positive-sign branch gives x - 2 = x - 6? Also impossible - so take x - 2 = -(x - 6): 2x = 8, x = 4. Geometrically, x = 4 is the midpoint of 2 and 6, directly below the midpoint of the two given points.

Why (B) is wrong: That is the first point itself.

Why (C) is wrong: 8 is beyond the second point - not symmetric.

Why (D) is wrong: 3 is not the average of 2 and 6.

Hinglish: *Dono given points ka y same hai (3), isliye equidistant point unke beech ke THEEK beech mein hoga: (2 + 6)/2 = 4. Equation se bhi x = 4 hi aata hai. Ye symmetry ka use hai - jab dono points same height par hon toh answer unka midpoint hota hai.*

MCQ 22 · Normal · Concept: Rectangle perimeter

22. The points (0, 0), (3, 0), (3, 4) and (0, 4) form a rectangle. Its perimeter is:

A. 12 units
B. 14 units
C. 24 units
D. 7 units

Correct Answer: (B)

Sides are 3 and 4. Perimeter = 2(3 + 4) = 14. (Its diagonal is 5, area 12.)

Why (A) is wrong: 12 is the area, not the perimeter.

Why (C) is wrong: Doubled the area by mistake.

Why (D) is wrong: Added only two sides.

Hinglish: *Perimeter = 2(length + width) = 2 x 7 = 14. 12 toh AREA hai - dono ko mix karna classic confusion hai.*

MCQ 23 · Normal · Concept: Order independence of distance

23. Statement: If the two points in a distance calculation are interchanged, the computed distance changes. Is the statement TRUE or FALSE, and why?

A. TRUE, because x and y swap places
B. FALSE, because the differences are squared, which erases the sign change
C. TRUE, because the formula is order-dependent
D. FALSE, because distance is always zero

Correct Answer: (B)

Swapping the points changes (x2 - x1) into (x1 - x2) = negative of it. But squaring makes both equal. Hence the distance is unchanged.

Why (A) is wrong: Coordinates do not swap - only the subtraction order flips.

Why (C) is wrong: The formula is order-independent.

Why (D) is wrong: Distance is zero only for identical points.

Hinglish: *Points ulta karne se differences ka sign ulta hota hai, par square laga-te-hi sign khatam. Isliye distance same rehta hai. Ye freedom hai - jis point ko pehle maano, jawab same.*

MCQ 24 · Normal · Concept: Axis membership

24. A point whose x-coordinate is 0 must lie:

A. on the x-axis
B. on the y-axis
C. in Quadrant II
D. at the origin

Correct Answer: (B)

x = 0 means zero sideways distance from the y-axis, so the point sits on the y-axis - at (0, y) for some y. It is the origin only if additionally y = 0.

Why (A) is wrong: On the x-axis it is y that equals 0.

Why (C) is wrong: Quadrant points have both coordinates non-zero.

Why (D) is wrong: Origin needs y = 0 as well.

Hinglish: *x = 0 matlab left-right displacement zero = seedha y-axis par. Origin tabhi banega jab y bhi 0 ho. x-axis wale options mein y = 0 hota hai - ulta mat karo.*

MCQ 25 · Normal · Concept: Equidistant on y-axis

25. The point (0, y) is equidistant from A(-3, 0) and B(5, 2). Then y equals:

A. 5
B. 2
C. 10
D. 4

Correct Answer: (A)

$9 + y^2 = 25 + (y - 2)^2$. Expand: $9 + y^2 = 25 + y^2 - 4y + 4$, so $9 = 29 - 4y$, giving $4y = 20$, $y = 5$. The point is (0, 5).

Why (B) is wrong: That is B's y-coordinate, not the solution.

Why (C) is wrong: Result of a sign slip in $(y - 2)^2$.

Why (D) is wrong: Arithmetic slip in moving terms.

Hinglish: *(0, y) rakho, dono squared distances barabar karo, y^2 cancel - linear equation: y = 5. Verify: (0,5) se A doori √34, B se bhi √34. Perfect.*

MCQ 26 · HOTS · Concept: Hypotenuse from coordinates

26. For the triangle with vertices (1, 1), (4, 1) and (1, 5), the hypotenuse has length:

A. 5 units
B. √41 units
C. 7 units
D. 6 units

Correct Answer: (A)

Legs along the axes directions: 4 - 1 = 3 and 5 - 1 = 4, meeting at the right angle (1, 1). Hypotenuse = √(9 + 16) = 5. The whole 3-4-5 triangle is just shifted to start at (1, 1).

Why (B) is wrong: √41 = √(16 + 25) comes from squaring wrong gaps.

Why (C) is wrong: 3 + 4 again.

Why (D) is wrong: Wrong leg difference.

Hinglish: *Origin par nahi hai triangle, (1, 1) par hai - par legs toh wahi 3 aur 4 hain! Translation se triple badalta nahi. Hypotenuse 5. √41 aata hai agar coordinates se seedha formula lagao bina right-angle pehchane.*

MCQ 27 · HOTS · Concept: Missing value on a line

27. If the points (2, 3), (4, k) and (6, 5) are collinear, then k equals:

A. 4

B. 3.5

C. 5

D. 2

Correct Answer: (A)

From (2, 3) to (6, 5) the total step is (+4, +2); halfway (at x = 4) the y-value must be the average: k = (3 + 5)/2 = 4. Indeed (4, 4) is the midpoint.

Why (B) is wrong: Averaged wrongly.

Why (C) is wrong: Copied the endpoint value.

Why (D) is wrong: No basis.

Hinglish: *Collinear matlab beech wala point line par. (2,3) se (6,5) ke beech ka x = 4 hai toh y bhi average hoga: (3+5)/2 = 4. Slope-check bhi karo: dono taraf slope 1/2. Clean.*

MCQ 28 · HOTS · Concept: Equally spaced chain

28. If (3, 2), (6, y) and (9, 8) are equally spaced collinear points, then y equals:

A. 5

B. 6

C. 4

D. 7

Correct Answer: (A)

Equal spacing means a constant step. x-step is 3 twice, so the y-step must also repeat: y = 2 + (8 - 2)/2 = 2 + 3 = 5. Check: steps (+3, +3) then (+3, +3).

Why (B) is wrong: Copied the x-step into y.

Why (C) is wrong: Halved wrongly.

Why (D) is wrong: No pattern use.

Hinglish: *Equally spaced = same step dono baar. x mein +3, +3 dikh raha hai, toh y mein bhi same jump: 2 se 8 tak total 6, half = 3, y = 5. Pattern dekho, equation ki zaroorat nahi.*

MCQ 29 · HOTS · Concept: Two-circle intersection

29. How many points are at distance 5 from BOTH (0, 0) and (8, 0)?

A. 0

B. 1

C. 2

D. Infinitely many

Correct Answer: (C)

The two circles (radius 5, centres 8 apart) overlap since $8 < 5 + 5 = 10$ and $8 > 0$. Two overlapping circles meet in exactly 2 points. By symmetry they are (4, 3) and (4, -3): check $16 + 9 = 25$. Correct.

Why (A) is wrong: Would need the circles separate ($8 > 10$).

Why (B) is wrong: Would need exactly touching ($8 = 10$).

Why (D) is wrong: That happens only for identical circles.

Hinglish: *Do circles socho - centres 8 door, radius 5-5. Wo do jagah kaatenge. Symmetry se x = 4 par, y = plus-minus 3 (kyunki 3-4-5 triple). Answer 2. Ek point tab hota jab circles ko touch karte (centres ki doori = radius ka sum).*

MCQ 30 · HOTS · Concept: Square from centre and vertex

30. A square has centre (2, 3) and one vertex at (5, 7). The area of the square is:

A. 50 sq units

B. 25 sq units

C. 100 sq units

D. $10\sqrt{2}$ sq units

Correct Answer: (A)

Centre-to-vertex distance = half the diagonal = $\sqrt{(3^2 + 4^2)} = 5$. So the full diagonal is 10, and area = diagonal$^2/2 = 100/2 = 50$. (Side = $10/\sqrt{2} = 5\sqrt{2}$, area 50.)

Why (B) is wrong: Used 5 as the side instead of half-diagonal.

Why (C) is wrong: Used 10 as the side.

Why (D) is wrong: That is a length, not an area.

Hinglish: *Centre se vertex tak = aadha diagonal = 5 (3-4-5 triple!). Poora diagonal 10, area = $d^2/2 = 50$. Ye diagonal-area relation square ke liye yaad rakho - side nikal-ghumane se fast hai.*

MCQ 31 · Normal · Concept: Circle membership

31. Which of these points does NOT lie on the circle $x^2 + y^2 = 25$?

A. (-3, -4)

B. (0, -5)

C. (4, -3)

D. (-4, -4)

Correct Answer: (D)

Check each: $9 + 16 = 25$ (on), $0 + 25 = 25$ (on), $16 + 9 = 25$ (on), but $16 + 16 = 32$ which is not 25. So (-4, -4) is off the circle (distance $4\sqrt{2}$ approx 5.66).

Why (A) is wrong: Both negatives still square to 9 and 16.

Why (B) is wrong: Zero coordinate is fine: $0 + 25 = 25$.

Why (C) is wrong: Same triple as A with signs flipped.

Hinglish: *Teeno pehle options 25 de rahe hain - circle par. (-4, -4) dekho: 32 nikla - bahar. Aise 'NOT' questions mein har option check karna padta hai, shortcuts nahi chalte.*

MCQ 32 · Normal · Concept: Distance expression

32. The distance of the point (a, b) from the origin equals:

A. a + b

B. $\sqrt{(a^2 + b^2)}$

C. $\sqrt{(a^2 - b^2)}$

D. $\sqrt{a} + \sqrt{b}$

Correct Answer: (B)

By the distance formula with the second point (0, 0): $\sqrt{((a - 0)^2 + (b - 0)^2)} = \sqrt{(a^2 + b^2)}$.

Why (A) is wrong: Adding lengths of legs never gives the hypotenuse.

Why (C) is wrong: A minus in the formula is fictitious.

Why (D) is wrong: Roots distribute over multiplication, not addition.

Hinglish: *Formula yaad karo - squares JUDTE hain, minus ya addition nahi. a + b isliye galat kyunki 3-4 wale case mein 7 deta hai jabki asli distance 5 hai.*

MCQ 33 · HOTS · Concept: Two possible y values

33. If the distance between A(1, 2) and B(4, y) is 5, the sum of ALL possible values of y is:

A. 4

B. 6

C. 8

D. 2

Correct Answer: (A)

$9 + (y - 2)^2 = 25$ gives $(y - 2)^2 = 16$, so y = 6 or y = -2. Both points (4, 6) and (4, -2) are at distance 5 from A. Sum = 6 + (-2) = 4.

Why (B) is wrong: Took only the positive root y = 6.

Why (C) is wrong: Added 6 + 6 or wrong pairing.

Why (D) is wrong: Took only the negative root.

Hinglish: *Do roots aayenge - circle A se guzar raha hai, vertical line x = 4 par do milaap. y = 6 aur y = -2. Question ne SUM manga hai = 4. Sirf ek value likhne wale question dhyan se nahi padhte.*

MCQ 34 · Normal · Concept: Double reflection

34. Reflecting (-2, -3) in the x-axis and then the image in the y-axis gives:

A. (2, 3)

B. (-2, 3)

C. (2, -3)

D. (-3, -2)

Correct Answer: (A)

x-axis reflection flips y: (-2, 3). Then the y-axis reflection flips x: (2, 3). Both flips together = sign of both coordinates reversed.

Why (B) is wrong: Only y flipped - one reflection.

Why (C) is wrong: Only x flipped - one reflection.

Why (D) is wrong: Swapped coordinates - reflections do not do that.

Hinglish: *Do mirror = dono sign flip. Seedha result (2, 3). Order matter nahi karta yahan - x pehle ya y pehle, same end point.*

MCQ 35 · Normal · Concept: Reflection on axis point

35. The reflection of the point (0, -7) in the x-axis lies:

A. on the x-axis
B. on the y-axis
C. in Quadrant III
D. at the origin

Correct Answer: (B)

Reflection flips y: (0, -7) becomes (0, 7). The x-coordinate is still 0, so the image sits on the y-axis. Points on the reflecting axis stay fixed, but this point was on the y-axis, not the x-axis, so it moves - along the y-axis.

Why (A) is wrong: Only points ON the x-axis stay put under x-reflection.

Why (C) is wrong: Quadrant III needs x negative too.

Why (D) is wrong: Origin only if both coordinates are zero.

Hinglish: *(0, -7) y-axis par baitha hai. x-axis mein reflect karne se wo y-axis ke upar jump karta hai: (0, 7). Axis par baithe point ko reflect karne ka rule: jis axis par hai wahi rehta hai (agar usi axis mein reflect karo), warna apni axis par hi travel karta hai.*

MCQ 36 · HOTS · Concept: Quarter-division point

36. P(-1, 4) and Q(7, -4) are joined. The point one quarter of the way from P to Q is:

A. (1, 2)
B. (3, 0)
C. (2, 1)
D. (1, 3)

Correct Answer: (A)

Total step Q - P = (8, -8). One quarter of the step = (2, -2). Add to P: (-1 + 2, 4 - 2) = (1, 2).

Why (B) is wrong: That is the midpoint (half-way), not the quarter point.

Why (C) is wrong: Step subtracted from P instead of added.

Why (D) is wrong: Wrong fraction of the step.

Hinglish: *Trisection wala idea hi hai, bas 1/4 ka step. Total jump (8, -8), uska chauthai (2, -2), P mein jodo: (1, 2). Option C aata hai jab log step ko P se MINUS kar dein.*

MCQ 37 · Normal · Concept: Sign logic

37. If a > 0 and b > 0, then the point (-a, -b) lies in:

A. Quadrant I
B. Quadrant II
C. Quadrant III
D. Quadrant IV

Correct Answer: (C)

-a and -b are both negative, giving the sign pattern (-, -), which is Quadrant III.

Why (A) is wrong: Needs both positive.

Why (B) is wrong: Needs x negative, y positive.

Why (D) is wrong: Needs x positive, y negative.

Hinglish: *Positive dono the, minus laga-te-hi dono negative = Q III. Signs hi sab kuch hain is chapter mein.*

MCQ 38 · Normal · Concept: Statement-based

38. Consider: (i) (x, y) equals (y, x) only when x = y. (ii) The distance between two points can be negative. (iii) Every point on the x-axis has y-coordinate 0. Which statements are TRUE?

A. (i) and (iii) only
B. (i) and (ii) only
C. (ii) and (iii) only
D. all three

Correct Answer: (A)

(i) is true: equal ordered pairs need equal coordinates. (ii) is false: distance is a length, always at least 0. (iii) is true by definition of the x-axis.

Why (B) is wrong: (ii) is never true.

Why (C) is wrong: (ii) is never true.

Why (D) is wrong: (ii) spoils the set.

Hinglish: *(i) sahi - ordered pair ka matlab hi coordinate-by-coordinate match hai. (ii) galat - distance kabhi negative nahi. (iii) sahi - x-axis ka definition. Toh (i) aur (iii).*

MCQ 39 · Normal · Concept: Symbolic horizontal distance

39. The distance between (a + b, c) and (a - b, c) is:

A. 2b units
B. 2|b| units
C. 2a units
D. |c| units

Correct Answer: (B)

Same y-coordinate c, so the distance is $|(a + b) - (a - b)| = |2b| = 2|b|$. The a-terms cancel.

Why (A) is wrong: Correct only when b is known positive; the safe answer carries the bar.

Why (C) is wrong: a cancels out completely.

Why (D) is wrong: c is irrelevant to a horizontal segment.

Hinglish: *y same hai, horizontal distance: $|a + b - a + b| = |2b|$. Bar isliye - b negative ho sakta hai, distance nahi ho sakti. a cancel ho gaya - uska koi role nahi.*

MCQ 40 · HOTS · Concept: Diagonal-step distance

40. Q = (3 + k, 4 + k) lies at distance $5\sqrt{2}$ from P(3, 4). Then k equals:

A. 5 only
B. -5 only
C. 5 or -5
D. no real value

Correct Answer: (C)

Distance = $\sqrt{(k^2 + k^2)} = \sqrt{(2k^2)} = |k|\sqrt{2}$. Setting $|k|\sqrt{2} = 5\sqrt{2}$ gives $|k| = 5$, so k = 5 or k = -5. Both directions along the diagonal line y = x work.

Why (A) is wrong: Missed the negative direction along the line.

Why (B) is wrong: Missed the positive direction.

Why (D) is wrong: Real solutions exist.

Hinglish: *Step (k, k) diagonal direction mein hai, distance $|k|\sqrt{2}$. $5\sqrt{2}$ se match karne par $|k| = 5$ - dono sign possible. Plus-minus bhoolna is question ki asli trap hai.*

MCQ 41 · Normal · Concept: Square perimeter

41. The perimeter of the square with vertices (-1, 2), (3, 2), (3, 6) and (-1, 6) is:

A. 16 units

B. 8 units

C. 32 units

D. 14 units

Correct Answer: (A)

Side = 3 - (-1) = 4. Perimeter = 4 x 4 = 16.

Why (B) is wrong: Two sides only.

Why (C) is wrong: Doubled for no reason.

Why (D) is wrong: Arithmetic slip.

Hinglish: *Side = x-gap = 4. Perimeter = 16. 8 sirf do sides, 32 toh area ke jaisa lagega ($4^2 = 16$ area actually) - perimeter 4 x side formula use karo.*

MCQ 42 · Normal · Concept: Equidistant + quadrant

42. A point is equidistant from both axes, lies in Quadrant II, and is at distance $4\sqrt{2}$ from the origin. The point is:

A. (4, 4)

B. (-4, 4)

C. (-4, -4)

D. $(4\sqrt{2}, -4\sqrt{2})$

Correct Answer: (B)

Equidistant from the axes with distance d from the origin forces $|x| = |y| = d/\sqrt{2} = 4$. Quadrant II demands x negative, y positive: (-4, 4).

Why (A) is wrong: Wrong quadrant (I).

Why (C) is wrong: Wrong quadrant (III).

Why (D) is wrong: Distances from axes would be $4\sqrt{2}$, not equal-to-origin relation.

Hinglish: *Distance $4\sqrt{2} = |x|\sqrt{2}$ (diagonal line) isliye $|x| = 4$. Q II: x minus, y plus. (-4, 4). Signs hi toh asli question the!*

MCQ 43 · HOTS · Concept: Circumcentre of right triangle

43. For the right triangle with vertices (0, 0), (6, 0) and (0, 8), the point equidistant from all three vertices is:

A. (3, 4)

B. (3, 5)

C. (4, 3)

D. (2, 3)

Correct Answer: (A)

Hypotenuse joins (6, 0) to (0, 8); its midpoint is (3, 4). In a right triangle the midpoint of the hypotenuse is equidistant from all three vertices (distance 5 here: 3-4-5 again).

Why (B) is wrong: Not the midpoint of the hypotenuse.

Why (C) is wrong: Swapped coordinates.

Why (D) is wrong: No geometric basis.

Hinglish: *Q42 wala theorem yaad karo - right triangle mein hypotenuse ka MIDPOINT hi teeno se equidistant. (6,0) aur (0,8) ka midpoint = (3, 4). Distance check: origin se 5, dono vertices se 5. Bang on.*

MCQ 44 · Normal · Concept: Circle through origin

44. A circle is centred at (3, 4) with radius 5. Which statement is TRUE?

A. The origin lies on the circle

B. The origin lies inside the circle

C. The origin lies outside the circle

D. The circle does not reach the axes

Correct Answer: (A)

Distance from (3, 4) to (0, 0) = $\sqrt{(9 + 16)} = 5$ = radius exactly, so the origin lies ON the circle. The circle touches both axes at (0, 4)... wait, distance from centre to the y-axis is 3 (less than 5) and to the x-axis is 4 (less than 5), so it crosses both axes.

Why (B) is wrong: Would need distance less than 5.

Why (C) is wrong: Would need distance more than 5.

Why (D) is wrong: The circle crosses both axes.

Hinglish: *Centre (3, 4) se origin ki doori = 5 = radius. Isliye origin circle PAR hai. Aur kyunki centre axis se 3 aur 4 door hai (5 se kam), circle dono axes ko kaat bhi raha hai. Option D bhi galat hai.*

MCQ 45 · Normal · Concept: Grid vs direct

45. On a unit grid, the shortest path from (2, 3) to (6, 7) moving only along grid lines has length 8. The straight-line distance is:

A. $4\sqrt{2}$ units

B. 8 units

C. $2\sqrt{2}$ units

D. 16 units

Correct Answer: (A)

Gaps: 6 - 2 = 4 and 7 - 3 = 4. Direct distance = $\sqrt{(16 + 16)} = \sqrt{32} = 4\sqrt{2}$ (approximately 5.66), versus 8 along the grid.

Why (B) is wrong: That is the grid distance.

Why (C) is wrong: Underestimated the root.

Why (D) is wrong: That is the squared distance 32 misread as 16 x ... no, 16 is not even the square here.

Hinglish: *4 aur 4 ka square: 32, root $4\sqrt{2}$. Grid par 8, seedha 5.66 - seedha rasta jeet gaya. $\sqrt{32}$ ko simplify karna mat bhoolo: $4\sqrt{2}$.*

MCQ 46 · Normal · Concept: Reflection preserves midpoints

46. A and B are two points and M is the midpoint of AB. A and B are reflected in the x-axis to A' and B'. Which statement is TRUE?

A. M' (the image of M) is the midpoint of A'B'

B. A'B' is longer than AB

C. M' does not exist on segment A'B'

D. The midpoint of A'B' is M itself, unchanged

Correct Answer: (A)

Reflection preserves distances and maps the segment onto the segment, so it maps the midpoint to the midpoint. The image M' is the midpoint of A'B'. (M itself moves unless it lies on the x-axis.)

Why (B) is wrong: Reflections preserve all distances.

Why (C) is wrong: The midpoint image always exists.

Why (D) is wrong: M moves under reflection unless it sits on the x-axis.

Hinglish: *Reflection poore segment ko utha ke shift karta hai - beech ka point beech ka hi rahega (image mein). AB ki length bhi same rehti hai - option B galat.*

MCQ 47 · Normal · Concept: Right triangle area, shifted

47. A(-1, -5), B(3, -5) and C(3, -2) form a right triangle. Its area is:

A. 6 sq units

B. 12 sq units

C. 5 sq units

D. 7 sq units

Correct Answer: (A)

AB is horizontal with length 4; BC is vertical with length 3. Area = (1/2)(4)(3) = 6. Negative coordinates do not change lengths or areas.

Why (B) is wrong: Forgot the 1/2.

Why (C) is wrong: Used the hypotenuse 5 as a leg.

Why (D) is wrong: Sum of legs instead of half-product.

Hinglish: *Sab negative zone mein hai par length toh gap hai: 4 aur 3. Area 6. Negative coordinates se dar mat - absolute gap lo.*

MCQ 48 · HOTS · Concept: Counting locus points

48. How many points satisfy BOTH |x| = 5 and |y| = 3?

A. 4

B. 2

C. 8

D. 1

Correct Answer: (A)

|x| = 5 gives x = 5 or -5; |y| = 3 gives y = 3 or -3. Combinations: (5, 3), (5, -3), (-5, 3), (-5, -3) - four points, one in each quadrant.

Why (B) is wrong: Counted only one sign combination.

Why (C) is wrong: Over-counted.

Why (D) is wrong: Missed the sign freedom.

Hinglish: *Do signs, do choices har ek ka = 2 x 2 = 4 points. Har quadrant mein ek. Ye counting-sawaal HOTS mein easy marks hai agar system se lo.*

MCQ 49 · Normal · Concept: Product of roots

49. If the point (x, 4) is at distance 5 from the origin, the PRODUCT of all possible values of x is:

A. -9
B. 9
C. 0
D. 3

Correct Answer: (A)

$x^2 + 16 = 25$ gives x = 3 or x = -3. Product = 3 x (-3) = -9.

Why (B) is wrong: Missed the negative root.

Why (C) is wrong: Sum would be 0, not product.

Why (D) is wrong: Single root only.

Hinglish: *3-4-5 triple phir! x = plus-minus 3. Product manga hai: 3 x (-3) = -9. Dono values nikalna zaroori tha, warna product hi galat.*

MCQ 50 · Normal · Concept: Nearest axis point

50. The point on the x-axis NEAREST to (4, 3) is:

A. (4, 0)
B. (0, 3)
C. (0, 0)
D. (3, 0)

Correct Answer: (A)

The shortest route from a point to a line is along the perpendicular. Dropping a perpendicular from (4, 3) to the x-axis lands at (4, 0), with distance 3.

Why (B) is wrong: (0, 3) is on the y-axis.

Why (C) is wrong: Origin is distance 5 away.

Why (D) is wrong: (3, 0) is distance √10 away.

Hinglish: *Line ka sabse nazdeeki point = perpendicular wala pair. (4, 3) se seedha niche = (4, 0), doori 3. y-coordinate hi doori thi - 3.*

MCQ 51 · Normal · Concept: Coinciding ordered pairs

51. The points (5, k) and (k, 5) coincide when k equals:

A. 0
B. 5
C. 5 or -5
D. any real number

Correct Answer: (B)

Two ordered pairs coincide when both coordinates match: 5 = k and k = 5 - the single condition k = 5.

Why (A) is wrong: k = 0 gives (5, 0) vs (0, 5) - different.

Why (C) is wrong: No distance or equation here to create two roots.

Why (D) is wrong: Only the diagonal line y = x hosts such coincidences.

Hinglish: *Ek hi point tab banega jab dono coordinate same: k = 5. Plus-minus wala confusion Q40 jaisa hai par wahan distance thi, yahan equality hai - yahan sirf ek value.*

MCQ 52 · Normal · Concept: 13-triple membership

52. Which of the following CANNOT be the coordinates of a point at distance 13 from the origin?

A. (5, 12)
B. (12, 5)
C. (6, 12)
D. (13, 0)

Correct Answer: (C)

Check squares against 169: 25 + 144 = 169 (yes), 144 + 25 = 169 (yes), 36 + 144 = 180 (no), 169 + 0 = 169 (yes). (6, 12) is outside the circle of radius 13.

Why (A) is wrong: Exactly the 5-12-13 triple.

Why (B) is wrong: Same triple swapped.

Why (D) is wrong: Axis point at full radius.

Hinglish: *$6^2 + 12^2 = 180$, 169 nahi - bahar. Baaki sab 13-triple family ke hain. Aise 'which CANNOT' questions mein har option ko 169 se compare karo.*

MCQ 53 · Normal · Concept: Symbolic diagonal distance

53. The distance between (t, t) and (t + 3, t + 7) is:

A. √58 units
B. 10 units
C. √52 units
D. 58 units

Correct Answer: (A)

Differences: (t + 3) - t = 3 and (t + 7) - t = 7. Distance = √(9 + 49) = √58. The t cancels - the distance is the same wherever the pair sits.

Why (B) is wrong: Confused with a non-existent triple.

Why (C) is wrong: Arithmetic slip: 9 + 49 is not 52.

Why (D) is wrong: Forgot the root.

Hinglish: *t cancel ho gaya dono difference mein. Bacha 3 aur 7: 9 + 49 = 58, root √58. Simplify nahi hota (58 = 2 x 29, koi square factor nahi). 58 ko root ke bahar mat likho - wo squared distance hai.*

MCQ 54 · HOTS · Concept: Classify quadrilateral

54. The points (1, 1), (4, 1), (6, 3), (3, 3) joined in order form a:

A. rectangle
B. parallelogram but not a rectangle
C. square
D. rhombus

Correct Answer: (B)

AB = 3 and CD = 3 (both horizontal); BC = √(4 + 4) = 2√2 and DA = 2√2 (both slanted with the same step (+2, +2)). Opposite sides equal and parallel: parallelogram. Diagonals: AC = √(25 + 4) = √29, BD = √(1 + 4) = √5, unequal - so not a rectangle. Sides are unequal too, so not a rhombus or square.

Why (A) is wrong: Diagonals unequal - fails the rectangle test.

Why (C) is wrong: Sides unequal.

Why (D) is wrong: Rhombus needs all sides equal.

Hinglish: *Opposite sides barabar (3, 3) aur (2√2, 2√2) - parallelogram. Diagonals check: √29 vs √5 - barabar nahi, toh rectangle nahi. Sides bhi unequal - rhombus/square nahi. Ye Q23-style classification ka MCQ avatar hai.*

MCQ 55 · HOTS · Concept: Ratio division point

55. P divides the segment from A(0, 0) to B(12, 8) in the ratio AP : PB = 1 : 3. Then P is:

A. (3, 2)

B. (4, 8/3)

C. (6, 4)

D. (9, 6)

Correct Answer: (A)

AP is one part out of four total parts, so P = A + (1/4)(B - A) = (12/4, 8/4) = (3, 2). Check AP = √13 and PB = √(81 + 36) = √117 = 3√13, ratio 1 : 3. Correct.

Why (B) is wrong: Took 1/3 of the way instead of 1/4.

Why (C) is wrong: That is the midpoint (ratio 1 : 1).

Why (D) is wrong: That is the 3 : 1 point.

Hinglish: *Ratio 1 : 3 matlab total 4 parts, P pehla part leke baitha hai. Step = B - A ka 1/4 = (3, 2). Verify: AP = √13, PB = 3√13 - ratio sahi. (6, 4) toh MIDPOINT hai (1:1 wala), ratio dhyan se padho!*

MCQ 56 · Normal · Concept: Vertical distance, negatives

56. The distance between (-2, 4) and (-2, -9) is:

A. 13 units

B. 11 units

C. √193 units

D. 5 units

Correct Answer: (A)

Same x, so distance = |4 - (-9)| = |4 + 9| = 13.

Why (B) is wrong: Dropped the sign of -9.

Why (C) is wrong: Used the full distance formula unnecessarily.

Why (D) is wrong: Subtracted 4 from 9 without the negative.

Hinglish: *x same - vertical line. 4 - (-9) = 13. Minus ke saath minus ka khel phir. 11 kaise? 4 + (-9) karke -7 ka absolute... nahi, 9 - 4 = 5? Option D wahi hai. Dono galat - sahi formula |y2 - y1|.*

MCQ 57 · Normal · Concept: Square from diagonal

57. A square has diagonal of length √72. Its area is:

A. 36 sq units

B. 72 sq units

C. 18 sq units

D. 6√2 sq units

Correct Answer: (A)

Area = diagonal^2/2 = 72/2 = 36. (Side = √72/√2 = √36 = 6.)

Why (B) is wrong: Forgot to halve d^2.

Why (C) is wrong: Halved twice.

Why (D) is wrong: That is the side, not the area.

Hinglish: *Square area = $d^2/2$ - seedha 36. Side 6, side-square 36. 72 likhne wale 1/2 bhool gaye.*

MCQ 58 · HOTS · Concept: Compare three distances

58. For the point (6, 8), which distance is the SMALLEST: to the x-axis, to the y-axis, or to the origin?

A. to the x-axis (8 units)

B. to the y-axis (6 units)

C. to the origin (10 units)

D. all are equal

Correct Answer: (B)

Distance to the x-axis = |y| = 8. Distance to the y-axis = |x| = 6. Distance to the origin = √(36 + 64) = 10. The comparison is 6 < 8 < 10, so the smallest is the 6-unit distance to the y-axis.

Why (A) is wrong: x-axis distance is 8, not 6.

Why (C) is wrong: Origin distance is always the largest of the three here.

Why (D) is wrong: 6, 8, 10 are distinct.

Hinglish: *Ruko, ye question galti-pakadne wala hai! x-axis se doori |y| = 8 hai, y-axis se doori |x| = 6 hai. Sabse chhoti 6 hai = y-axis se doori. Asli seekh: x-axis se doori y-coordinate hoti hai, y-axis se doori x-coordinate - ULTA mat karna, warna 6 aur 8 ki jagah badal jaayegi.*

MCQ 59 · Normal · Concept: Assertion-reason axis

59. Assertion (A): The point (0, 4) lies on the y-axis. Reason (R): Every point whose x-coordinate is 0 lies on the y-axis.

A. Both A and R are true and R explains A

B. Both A and R are true but R does not explain A

C. A is true, R is false

D. A is false, R is true

Correct Answer: (A)

(0, 4) has x = 0, which is exactly the reason R states for lying on the y-axis. Both are true and R is the direct justification of A.

Why (B) is wrong: The connection is direct, not incidental.

Why (C) is wrong: R is the standard definition.

Why (D) is wrong: A is clearly true.

Hinglish: *(0, 4) ka x zero hai - aur y-axis ka definition bhi yahi hai. R hi A ka logic hai. Straightforward A-R: option A.*

MCQ 60 · HOTS · Concept: 9-12-15 triangle

60. The perimeter of the triangle with vertices (0, 0), (9, 0) and (0, 12) is:

A. 36 units

B. 30 units

C. 42 units

D. 39 units

Correct Answer: (A)

Legs 9 and 12; hypotenuse = √(81 + 144) = √225 = 15 (the 9-12-15 triple = 3 x (3-4-5)). Perimeter = 9 + 12 + 15 = 36.

Why (B) is wrong: Used 5-12-13 by mistake.

Why (C) is wrong: Overshoot on the hypotenuse.

Why (D) is wrong: Partial sum.

Hinglish: *9 aur 12 ko dekho: dono 3 se divide hote hain - 3-4-5 ka scaled version, hypotenuse 15. Perimeter 36. Triples ka scaling trick bade numbers ko chhota kar deta hai.*

1.5 Chapter Formula Sheet

Every working formula of Chapter 1 in one table. Read the conditions column carefully - most wrong answers in exams are correct formulas applied in illegal situations.

Formula	Meaning and when to use	Conditions	Tiny example	Common mistake
Distance from O = $\sqrt{(x^2 + y^2)}$	Length of the segment joining any point to the origin; use whenever 'how far from O' appears	Any real x, y	(6, 8): $\sqrt{(36 + 64)} = 10$	Writing x + y instead of the square-root sum
$d = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$	Distance between ANY two points; the master formula	Works for all real coordinates, any order of the points	(1, 2) to (4, 6): $\sqrt{(9 + 16)} = 5$	Taking the root separately on each difference
Horizontal distance = $\lvert x_2 - x_1 \rvert$	Two points with the same y-coordinate	$y_1 = y_2$	(2, 5) to (9, 5): 7	Forgetting the absolute value when $x_2 < x_1$
Vertical distance = $\lvert y_2 - y_1 \rvert$	Two points with the same x-coordinate	$x_1 = x_2$	(0, -4) to (0, 6): 10	Computing 6 - (-4) as 2 instead of 10
Midpoint M = $((x_1 + x_2)/2, (y_1 + y_2)/2)$	The point that bisects a segment; also proves diagonals bisect	Always valid for a segment	(-3, 8) and (5, -2) give (1, 3)	Subtracting instead of adding, or forgetting to halve
Reverse midpoint: B = $(2m_1 - x_1, 2m_2 - y_1)$	One endpoint known with the midpoint; find the other endpoint	Midpoint given	M(2, -1), A(-4, 3) give B(8, -5)	Adding the coordinates instead of subtracting
Trisection: P = A + (B - A)/3	Points cutting a segment into three equal parts	P closer to A; Q = A + 2(B - A)/3	A(4, 7), B(16, -2) give P(8, 4), Q(12, 1)	Using half the gap (that gives the midpoint)
Reflection in x-axis: (x, y) → (x, -y)	Mirror image across the horizontal axis	Only y flips	(4, -3) gives (4, 3)	Flipping x as well (that is a double reflection)
Reflection in y-axis: (x, y) → (-x, y)	Mirror image across the vertical axis	Only x flips	(-3, -5) gives (3, -5)	Flipping y instead of x

Formula	Meaning and when to use	Conditions	Tiny example	Common mistake
Circle test: on $x^2 + y^2 = r^2$; inside if $x^2 + y^2 < r^2$; outside if $x^2 + y^2 > r^2$	Classify a point against a circle centred at O	Centre must be the origin	(4, 2) with r = 5: 20 < 25, inside	Comparing roots instead of squares
Collinearity: largest distance = sum of other two	Three points on one line	Use EXACT values, no early rounding	5 + 5 = 10 for (0,0), (3,4), (6,8)	Comparing rounded decimals in near-miss cases
Perimeter of polygon = sum of side distances	Vertices given; add up the side lengths	Compute each side exactly	8 + 5 + 8 + 5 = 26 for a 8 x 5 rectangle	Using a wrong pair of coordinates for one side
Area of right triangle = (1/2) x leg x leg	Right angle visible (legs axis-parallel or via converse Pythagoras)	Legs must be the two sides forming the right angle	(0,0), (8,0), (0,6): area 24	Forgetting the 1/2, or using the hypotenuse as a leg

1.6 Last-Minute Cheat Sheet

> **30-second revision card - Chapter 1**
> SIGNS: Q I (+,+), Q II (-,+), Q III (-,-), Q IV (+,-). x = 0 means y-axis, y = 0 means x-axis. DISTANCES: same y, use $|x_2-x_1|$; same x, use $|y_2-y_1|$; otherwise $\sqrt{(dx^2+dy^2)}$. From origin: $\sqrt{(x^2+y^2)}$. TRIPLES: 3-4-5, 5-12-13, 8-15-17, 7-24-25 and their multiples. MIDPOINT = averages; reverse = double minus. Reflection: x-axis flips y, y-axis flips x. CIRCLE: compare x^2+y^2 with r^2. COLLINEAR: biggest = sum of other two. (a,b) = (b,a) only if a = b.

- **If you see two points sharing a coordinate** → single-difference shortcut, no square root.
- **If you see 'find k' with a distance** → absolute value opens BOTH ways; expect two answers.
- **If you see 'inside / on / outside the circle'** → compare $x^2 + y^2$ with r^2, never roots.
- **If you see 'midpoint of ... is M, find the other end'** → double the midpoint, subtract the known end.
- **If you see 'reflected in the axis'** → flip exactly ONE sign; two reflections flip both.
- **If you see 'without plotting, same line?'** → largest distance equals sum of the other two (exact values only).
- **If you see legs on the axes of a triangle** → right angle at the shared axis point; area = half product of legs.
- **If a question shifts a 3-4-5-style shape away from the origin** → the triple survives translation; just use gaps.

1.7 Mistake Buster - DON'T MAKE THESE MISTAKES

1. **Coordinate-slot swap.** Distance from the y-axis is the x-coordinate, and from the x-axis is the y-coordinate. Writing them the other way round silently ruins the answer. Fix: say aloud 'x talks about the y-axis' while practising.

2. **Sign collapse in subtraction.** 6 - (-3) = 9, not 3. Every vertical-distance error trace leads here. Fix: rewrite a - (negative) as a + positive BEFORE computing.
3. **Root-splitting the formula.** $\sqrt{(dx^2 + dy^2)}$ is NOT $\sqrt{dx} + \sqrt{dy}$ and NOT dx + dy. Fix: chant 'square, square, add, then root - once'.
4. **Losing a root in 'find k' problems.** |4 - k| = 7 has two solutions. Fix: always write the two cases explicitly.
5. **Comparing rounded roots for collinearity.** $\sqrt{5} + 5$ vs $2\sqrt{13}$ differ by less than 0.03; decimals mislead. Fix: compare squared values exactly.
6. **Mixing up (x, y) and (y, x).** They coincide only on the line y = x. Fix: never 'sort' coordinates alphabetically or by size.
7. **Distance written as negative.** Distances are never negative; a negative coordinate is a DIRECTION. Fix: absolute value before reporting any length.
8. **Midpoint halving forgotten.** (x_1+x_2, y_1+y_2) is the double midpoint, not the midpoint. Fix: divide by 2 as a separate written step.
9. **Reflection in the wrong variable.** x-axis reflection flips y. Students flip x because the axis is named x. Fix: 'the axis you reflect in stays fixed - the other one flips'.
10. **Using the hypotenuse as a leg in area.** Area = half product of the two LEGS (the sides touching the right angle). Fix: locate the right angle first, then pick its two arms.

1.8 Exam Strategy for Chapter 1

Order of attack. Begin with identification and axis-distance questions (they take under a minute each and settle your nerves). Then do standard distance-formula questions. Keep classification questions (prove square/parallelogram) and multi-part HOTS for the end, when you can give them the calm they need.

Where marks are lost. In this chapter, examiners report three leak-points: sign errors under time pressure (a 2-mark question gone), single-root answers in 'find k' questions (half marks gone), and unverified classifications (method marks gone even when the answer is right). The antidote for all three is the same: write the check line - verify one distance, verify both roots, verify the defining property.

Formulas most frequently tested. The distance formula (appears in almost every question), the parallel-to-axis shortcuts, the midpoint formula in both directions (find midpoint, find missing endpoint), reflection rules, and the $x^2 + y^2$ versus r^2 comparison. The 3-4-5 family of triples is effectively a free speed boost in half of all numerical questions.

Concepts that get mixed. Distance with midpoint (find endpoints from midpoints), distance with circle membership (inside/outside tests), reflections chained with distance (prove preservation), and collinearity with equal spacing (chain-step pattern). When two concepts combine, handle them in labelled mini-steps rather than one mega-computation.

Safe shortcuts. Using recognised triples (only when the numbers match exactly), shifting both points by a common vector before computing distance, and comparing squared distances instead of roots. **Unsafe shortcuts.** Rounding roots early to compare, guessing the quadrant from a sketch drawn freehand, dropping the absolute value bars 'because the answer looks positive', and applying the midpoint formula when the question actually describes a trisection point.

How to check an answer. Distance: recompute with the points interchanged - the result must be identical. Midpoint: re-average your answer with the other endpoint - you must recover the given midpoint. Reflection: check one preserved distance. Circle: recompute $x^2 + y^2$. Two minutes of checking routinely saves four to six marks.

1.9 Chapter Completion Checklist

Tick each box only when you can honestly say DONE:

- [ ] Every concept covered: axes, origin, ordered pairs, quadrants, axis points, distances, distance formula, reflections, midpoint, trisection, collinearity, circle membership
- [ ] Every formula covered: 13 formulas in Section 1.5, each with conditions and a micro-example
- [ ] 50 advanced questions completed (Q1-Q50), difficulty ladder respected
- [ ] Every question has a full step-by-step solution
- [ ] Every solution has a Hinglish explanation
- [ ] 60 MCQs completed with answers
- [ ] Every MCQ has an explanation plus trap analysis
- [ ] Formula sheet completed
- [ ] Last-minute cheat sheet completed
- [ ] Mistake buster completed (10 named mistakes)
- [ ] Exam strategy completed